

FINAL YEAR PROJECT REPORT

Project title:

**Design and development of a multi-agent system for queue
waiting-time estimation from video streams**

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Dedications

To my parents,

No dedication could ever truly reflect my unending love and gratitude for all you've done for my education and well-being. May you recognize in this simple work the fulfillment of your dreams and the reward of your many sacrifices.

To my sister,

Your affection, support, and constant encouragement have been an immense source of motivation throughout my journey. Your belief in me helped me overcome challenges and stay focused on my goals. I wish you a life filled with success and happiness.

To my family,

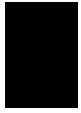
Your presence and unconditional love are my greatest treasures. Thank you for everything. You will forever remain my inexhaustible source of support, love, and comfort.

To all my friends,

My second family who made my journey an indescribable experience, and who have always encouraged me and believed in me. I wish you a future filled with achievement and joy.

To all of you

I dedicate this work.



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List of Abbreviations and Symbols

The following symbols and abbreviations are used throughout the report.

Symbols

λ	Arrival rate.
μ	Service rate.
W	Expected wait time.
L	Number of people currently in the queue zone.

Abbreviations

API

Application Programming Interface.

BFF

Backend for Frontend.

ByteTrack

Multi-object tracking algorithm used to preserve identity continuity.

CSV

Comma-Separated Values.

ESG

Environmental, Social, and Governance.

EMEA

Europe, Middle East, and Africa.

FastAPI

Python web framework used for the backend API.

HTTP

Hypertext Transfer Protocol.

KPI

Key Performance Indicator.

MJPEG

Motion JPEG.

M/M/1

Single-server queueing model with Markovian arrivals and service times.

n8n

Workflow automation platform used for webhook routing and notifications.

ONVIF

Open Network Video Interface Forum.

OpenCV

Open source computer vision library.

Pydantic

Data validation library used with FastAPI.

REST

Representational State Transfer.

ROI

Return on Investment.

RTSP

Real-Time Streaming Protocol.

SQLite

Lightweight embedded SQL database engine.

TCP

Transmission Control Protocol.

UDP

User Datagram Protocol.

UI

User Interface.

WebRTC

Web Real-Time Communication.

WHEP

WebRTC HTTP Egress Protocol.

WS-Discovery

Web Services Discovery protocol used for ONVIF camera discovery.

YOLO

You Only Look Once.



General Introduction

Service organizations are under constant pressure to reduce waiting time and improve service quality. Supermarkets, banks, and service centers all face the same problem: queues change quickly, are difficult to monitor by hand, and react strongly to demand fluctuations.

In practice, customers often join a queue without knowing how long they will wait, while managers and operational staff rarely have reliable real-time indicators for adjusting resources quickly. Manual observation is subjective, and simple counting tools are not enough to estimate waiting time in a meaningful way.

This end-of-studies project addresses that gap through the design and development of a real-time queue monitoring system based on computer vision and queueing theory. The proposed system detects and tracks people in video streams, analyzes occupancy inside configurable queue zones, estimates arrival and service rates, computes waiting time, and exposes metrics that can be used operationally.

It combines classical queueing theory with computer-vision detection and tracking techniques [1], [2] to provide actionable waiting-time estimates.

This report is organized as follows. Chapter 1 presents the project context, host organization, problem statement, and objectives. Chapter 2 defines the requirements analysis and the main architecture choices. Chapters 3 to 6 follow the implementation sprint by sprint, from perception and queue analytics to frontend integration and operational automation. The general conclusion closes the report and presents future work directions.

Context of the Project

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1.1 Introduction

This chapter situates the Queue Wait-Time Estimation System in its operational and organizational context. It explains why queue monitoring remains important for service organizations and why a complete technical solution is needed instead of isolated counting tools.

It also sets the baseline used throughout the report: the collaboration with DataDoIt, the problem statement, the comparison with existing approaches, the solution scope, and the Scrum organization of the work. These points matter because the later technical chapters rely on them.

1.2 Host Organization

1.2.1 DataDoIt

This end-of-studies project was carried out with DataDoIt, a data-driven sustainability consulting firm headquartered at Elghazal Technopark in Tunis, Tunisia. DataDoIt works on projects that connect data science and sustainability for governments, municipalities, and enterprises across the EMEA region. Its services cover decarbonization strategy, ESG reporting, smart resource analytics, and circular economy implementation. The company uses the motto Measure It, Optimize It, Do It to describe a working method based on analysis, planning, and deployment. At the time of the project, it had delivered more than 15 active projects across 6 sectors, with a reported average ROI of 2.3x and a 94% client retention rate.

The collaboration provided a concrete setting for work centered on data collection, real-time monitoring, and automated analytics, and these concerns directly shaped the design and scope of the project, with the DataDoIt company logo shown in Figure 1.1.



Figure 1.1: DataDoIt company logo

The collaboration perimeter that framed the project is summarized in Table 1.1.

Dimension	Scope in this project
Business context	Queue management in service scenarios where waiting time and staffing decisions are operationally sensitive.
Engineering constraints	Multi-source video ingestion, variable visual conditions, low-latency monitoring, and integration with automation channels.
Expected outputs	Actionable queue indicators, wait-time estimation, and alert-ready payloads for external workflows.
Validation mode	Iterative sprint reviews using implementation evidence from backend modules, API contracts, frontend behavior, and automation tests.

Table 1.1: Collaboration perimeter with DataDoIt

1.2.2 Sectors of Activity

DataDoIt delivers sustainability solutions across six mission-critical sectors where data-driven environmental performance has direct operational and financial implications:

- government and public sector institutions seeking to translate climate policy into measurable action,
- smart cities and municipalities requiring integrated urban resilience and climate adaptation systems,

- industrial and manufacturing operations targeting Scope 1-3 carbon footprint reduction and supply chain efficiency,
- real estate and infrastructure organizations pursuing green certification and climate-resilient asset development,
- financial services firms navigating ESG compliance and green finance opportunities,
- energy and utilities providers accelerating the transition toward renewable integration and grid modernization.

1.3 Presentation of the Project

1.3.1 Project Context

Manual queue supervision is often subjective, hard to scale, and dependent on continuous attention. In many settings, teams react only after congestion appears instead of anticipating it with measurable indicators.

This project addresses that gap by turning camera streams into decision-oriented telemetry. The target output is not limited to occupancy counts, but includes people in zone, arrival rate, service rate, adaptive wait-time estimates, and alert states that can be consumed by monitoring and automation layers.

1.3.2 Problem Statement

Current operational practices and basic counting tools still have recurring limitations:

- they provide limited anticipation of congestion,
- they rarely estimate waiting time explicitly,
- they provide weak visibility into queue stability during sudden demand spikes,

- they are often disconnected from automated response workflows.

As a result, interventions such as staff reallocation, lane opening, or escalation can be delayed or triggered on incomplete signals. The problem is therefore both analytical and integration-related: it is not enough to produce robust indicators; they also need to be delivered through actionable channels, and Table 1.2 maps these pain points to system-level requirements.

Observed pain point	Operational impact	Required system capability
Manual visual estimation is inconsistent across shifts.	Queue state interpretation varies between staff members, reducing decision reliability.	Standardized per-feed metrics with explicit formulas and stable update cadence.
Occupancy-only monitoring is descriptive but not predictive.	Teams react late because no forward-looking wait-time signal is available.	Continuous estimation of λ , μ , and expected wait time W .
Queue behavior under load spikes is hard to interpret.	Teams may overreact or underreact when service and arrivals fluctuate quickly.	Adaptive wait-time formula that responds gracefully to changing queue conditions.
Alerting tools are not integrated with runtime analytics.	Escalation depends on manual relay, increasing response latency.	Automated webhook dispatch and workflow-based notification routing.

Table 1.2: From operational pain points to system-level requirements

1.3.3 Study of Existing Approaches

Several computer-vision frameworks provide the basic building blocks for person detection and tracking. In parallel, queueing theory gives a formal way to describe service-system behavior and relate arrival rate, service rate, and expected waiting time [1].

In practice, existing commercial and open-source tools cover only parts of the

queue-management problem. They fall into three broad families: queue management platforms, sensor-based analytics, and vision-based monitoring frameworks.

1.3.3.1 Queue Management Platforms

Platforms such as Waitwhile [3] and Qminder [4] focus on active queue management. They let customers join virtual queues through kiosks, QR codes, or web links, and they notify staff through SMS or dashboards. Their strength is customer-facing experience: reduced perceived waiting, appointment scheduling, and service routing. However, they require explicit customer interaction (check-in) and do not observe the physical queue directly. They cannot estimate wait time from the scene itself, and they do not detect congestion unless a customer has already registered.

1.3.3.2 Sensor-Based Analytics

RetailNext [5] and Xovis [6] represent the sensor-based analytics family. RetailNext uses ceiling-mounted AI sensors to track foot traffic, measure conversion rates, and optimize staffing in retail environments. Xovis uses 3D optical sensors for precise people flow and queue length measurement, particularly in airports and large retail spaces. Both provide high-accuracy occupancy and dwell-time data, but they require proprietary hardware installations and are primarily designed for large-scale enterprise deployments. They observe the queue passively but do not typically integrate with external automation workflows or provide open APIs for custom alert routing.

1.3.3.3 Vision-Based Monitoring Frameworks

Open-source tools built on YOLO detection and tracking, such as those provided by Ultralytics [2] and Roboflow [7], offer the computer-vision building blocks needed for queue monitoring: person detection, tracking, and zone-based counting. These frameworks are flexible and can be adapted to specific use cases. However, they

typically stop at detection and counting. They do not include queue-theoretic estimation of wait time, stability classification, or integration with notification workflows. The gap between raw detection output and actionable operational metrics is left to the developer.

1.3.3.4 Positioning of the Proposed Solution

The proposed solution is positioned against the main existing approach families in Table 1.3.

Approach family	Strengths	Typical limitation
Queue management platforms (Waitwhile, Qminder)	Customer-facing experience, virtual queues, appointment scheduling.	Require explicit check-in; do not observe physical queues or estimate wait time from the scene.
Sensor-based analytics (RetailNext, Xovis)	High-accuracy occupancy and flow data from dedicated 3D sensors.	Proprietary hardware; limited custom automation; enterprise-scale pricing.
Vision-based detection frameworks (Ultralytics, Roboflow)	Flexible open-source detection and tracking building blocks.	Stop at counting; no queue-theoretic estimation, stability classification, or alert integration.
Manual or occupancy-only monitoring	Easy to deploy and interpret.	Weak predictive value and high subjectivity.
Proposed integrated pipeline	Links perception, queue analytics, stability handling, web monitoring, and automation in one modular architecture.	Requires careful integration and validation across multiple technical domains.

Table 1.3: Positioning of the proposed solution against existing approaches

The proposed solution fills the gap between these families by combining open-source vision-based detection with queue-theoretic analytics, a web monitoring dashboard, and automated alert routing, all in one modular pipeline that does not require proprietary hardware or explicit customer interaction.

1.3.4 Proposed Solution

The system combines perception, analytics, and integration in one modular pipeline:

1. person detection and multi-object tracking from camera feeds,
2. polygon-based queue-zone filtering,

3. estimation of λ , μ , and expected wait time,
4. adaptive queue-state metrics that remain interpretable under varying load,
5. real-time web supervision through tiered live transport (WebRTC primary, MJPEG fallback),
6. automation through n8n webhooks and Telegram alerts.

This architecture converts visual observations into operational actions while keeping each layer independently testable. Detection and tracking modules can evolve without rewriting queue formulas, and the web and automation channels can evolve without changing the core metric contract.

1.3.5 Objectives of the Project

The project objectives are to:

1. detect and track people in real time,
2. count people in configurable queue zones,
3. estimate arrival and service rates,
4. compute expected wait time,
5. estimate wait time adaptively for operational reliability,
6. expose metrics through web interfaces and automation workflows.

To keep the implementation traceable, the objectives are mapped to delivery modules and observable outputs in Table 1.4.

Objective ID	Objective	Primary module	Observable output
OBJ-01	Real-time detection and tracking	Module A	Stable in-frame detections with persistent tracker IDs.
OBJ-02	Queue-zone counting	Module A	Per-frame in-zone people count and zone-aware overlays.
OBJ-03	Queue metric estimation (λ, μ, W)	Module A	Continuous metric payloads from queue analysis pipeline.
OBJ-04	Adaptive wait-time estimation	Module A	Compute wait time with adaptive logic that responds to queue load.
OBJ-05	Web monitoring	Module B	Live dashboard with feed controls, KPIs, and stream views.
OBJ-06	Automation integration	Module C	Webhook-based alert routing through n8n and Telegram.

Table 1.4: Objective-to-module traceability

1.3.6 Project Scope and Boundaries

The implementation scope covers backend perception and analytics modules, API and runtime integration, web monitoring, and external workflow automation. The report focuses on this web-first operational architecture.

All implementation details and validation narratives are presented through the backend, web, and automation tracks to keep the scope consistent.

1.3.7 Target Audience

The system is intended for three stakeholder groups with complementary goals:

- **Administrators:** manage manager accounts and access governance.
- **Managers:** configure operational settings, monitor live queue status, react to alerts, and use indicators for staffing and service-quality planning.
- **End users:** benefit indirectly from reduced waiting times and smoother service

flow.

This separation is important because each group consumes different levels of abstraction, from per-feed live events to aggregated performance trends.

1.4 Project Management Methodology

1.4.1 Choice of Methodology: Agile

An Agile approach was selected to deliver functional increments quickly while reducing integration risks early. The project combines multiple technical domains, including computer vision, queue analytics, backend APIs, frontend workflows, and automation, and each one has uncertainties that are difficult to resolve in a single delivery cycle.

Iterative delivery made it possible to validate contracts progressively: first perception outputs, then analytical reliability, then API/runtime integration, then web usability, and finally workflow automation and hardening.

1.4.2 SCRUM

Development followed sprint-based Scrum planning with iterative validation. For report readability and implementation traceability, the work is presented as four consolidated sprint chapters that are grouped into three logical modules.

The Scrum methodology used to organize the project is illustrated in Figure 1.2.

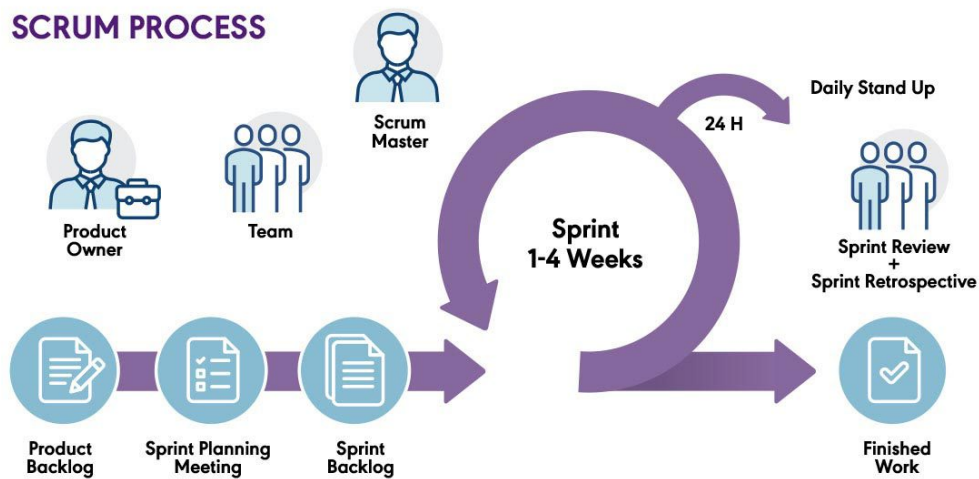


Figure 1.2: Scrum process used to organize the project.

1.4.2.1 Module A: Backend Core (Sprints 1–2)

This module established the vision, analytics, and backend service backbone:

- Sprint 1: detection, tracking, zone filtering, and queue metrics estimation,
- Sprint 2: camera connectivity, backend API exposure, and live transport.

1.4.2.2 Module B: Web Experience (Sprint 3)

This module focused on the web supervision experience:

- Sprint 3: web design system, frontend scaffolding, live backend integration, and dashboard monitoring.

1.4.2.3 Module C: Automation and Finalization (Sprint 4)

This module covered automation, evaluation, and project closure:

- Sprint 4: n8n automation, Telegram alerting, evaluation, optimization, and project consolidation.

The Scrum module mapping used in this report is summarized in Table 1.5.

Module	Sprints	Main focus	Validation perspective
Module A: Backend Core	1 to 2	Perception pipeline, queue analytics, camera connectivity, API services, live transport	Perception correctness, metric contracts, and API reliability.
Module B: Web Experience	3	Frontend design system, dashboard scaffolding, live integration, monitoring workflows	Usability, live-rendering stability, and role-based navigation.
Module C: Automation and Finalization	4	n8n alerting, Telegram notifications, evaluation, optimization, and fault tolerance	End-to-end operability and delivery readiness.

Table 1.5: Scrum module mapping used in this report

1.4.3 Sprint Planning

The project followed a sprint-based Scrum structure to keep the work incremental and traceable from requirements to implementation. Each sprint focused on one coherent part of the system, and the overall plan was adjusted as the internship progressed, as detailed in Table 1.6.

Sprint	Objectives	Start Date	End Date	User Story IDs
Sprint 1	• Perception and queue analytics • Detection, tracking, and zone filtering • Wait-time estimation and stability handling	16/02/2026	14/03/2026	US1.1–US1.9
Sprint 2	• Camera connectivity and discovery • Backend API exposure • Live transport and feed management	15/03/2026	11/04/2026	US2.1–US2.4
Sprint 3	• Web design system and frontend scaffolding • Live backend integration • Dashboard monitoring and role-based navigation	12/04/2026	09/05/2026	US3.1–US3.9
Sprint 4	• n8n automation and Telegram alerting • Testing and verification • Runtime tuning and fault tolerance	10/05/2026	06/06/2026	US4.1–US4.7

Table 1.6: Sprint planning used in the report

The detailed user stories for each sprint are presented in the backlog tables in Chapters 3 to 6.

1.5 Conclusion

This chapter set the project context, formalized the addressed operational problem, and justified the selected delivery strategy. Chapter 2 builds on this baseline by defining the requirements, architecture constraints, and analysis artifacts used to guide implementation decisions.

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2.1 Introduction

This chapter turns the project objective into the requirements that guided the implementation. It identifies the actors, lists the functions and quality constraints they need, and shows how those requirements are spread across the implementation plan.

The requirements are tied to the queue-monitoring pipeline used throughout the project: video ingestion, person detection and tracking, queue-metric estimation, dashboard delivery, authentication, and alert automation. This connection matters because the later chapters follow the same path from requirement to implementation, including the access layer used for manager onboarding and administrator governance.

2.2 Requirements Specification

2.2.1 Identification of Actors

The platform involves two in-system actors and one external operational actor. This separation clarifies which actions are executed inside the software and which actions happen on site.

The actors are:

- **Administrator:** manages manager accounts (create, update, delete) and controls access governance.
- **Manager:** configures sources and thresholds, monitors live queues, reacts to alerts, and reviews performance indicators.
- **On-site Operator (external actor):** receives operational instructions from the manager and applies field actions (for example opening a new queue line or redirecting customer flow), then confirms completion.

This split makes the functional scope easier to explain and validate: access-control

and monitoring actions remain in-system, while physical queue intervention is explicitly modeled as an external operational action.

2.2.2 Authentication and Access Governance

The platform includes a lightweight identity layer that restricts dashboard access to authenticated users, enabling managers to register, sign in, refresh sessions, and log out while allowing administrators to manage manager accounts and review access-related actions. Although the detailed backend and frontend implementation is deferred to the sprint chapters, these requirements are captured here because access boundaries form part of the core system behavior, as shown in Table 2.1.

ID	Requirement	Priority	Verification
AR-01	Allow a manager to register, sign in, refresh, and log out through secure cookies.	High	Validate register, login, refresh, logout, and session checks.
AR-02	Allow the administrator to create, update, deactivate, and reset manager accounts.	High	Validate account-governance endpoints and audit events.
AR-03	Enforce password policy, lockout after repeated failures, and role-based protection on operational endpoints.	High	Validate failed-login counters, lockout behavior, and unauthorized access.
AR-04	Record authentication and account-governance actions for later traceability.	Medium	Validate audit-log entries during register and login flows.

Table 2.1: Authentication and access-control requirements

2.2.3 Functional Requirements

At a high level, the system captures a feed, counts people, estimates queue behavior, updates the dashboard, and sends alerts when needed. From that workflow, the system must provide the following capabilities:

- real-time person detection and tracking,
- configurable queue zone definition,
- estimation of arrival rate (λ), service rate (μ), and wait time (W),

- adaptive wait-time estimation with robust fallback behavior,
- real-time web dashboard updates and alerts,
- IP camera integration through complementary ONVIF discovery and RTSP streaming,
- automated webhook and Telegram notifications,
- secure manager sign-up, sign-in, and role-based access control,
- historical data logging and analytics.

These requirements define the minimum end-to-end behavior expected from the platform and are formalized in the traceability matrix shown in Table 2.2.

ID	Requirement	Priority	Verification
FR-01	Detect people in live or recorded streams in near real time.	High	Validate detections and overlays in the perception layer.
FR-02	Preserve identity continuity across frames for queue analysis.	High	Validate tracker ID continuity in runtime scenarios.
FR-03	Restrict counting to configurable polygon queue zones.	High	Validate polygon inclusion in the zone-management layer.
FR-04	Compute λ , μ , and expected wait time from event windows.	High	Validate queue formulas and stability handling in analytics.
FR-05	Compute wait time with adaptive logic that switches between precise and conservative estimates.	High	Validate formula branches and fallback behavior.
FR-06	Expose feed control and live updates through API and websocket channels.	High	Validate endpoint and runtime behavior in API tests.
FR-07	Dispatch alert payloads to automation workflows.	Medium	Validate webhook dispatch and n8n branching.
FR-08	Persist operational metrics and metadata for analysis and auditing.	Medium	Validate persistence behavior in runtime and database checks.
FR-09	Support manager registration, sign-in, refresh, and logout through secure cookies.	High	Validate register, login, refresh, logout, and session checks.
FR-10	Protect administrative and operational APIs with role-based access control and audit logging.	High	Validate unauthorized access paths and audit events.

Table 2.2: Functional requirement traceability matrix

2.2.4 Non-functional Requirements

In addition to functional behavior, the platform must satisfy quality constraints that keep it usable in a live monitoring environment. Because the project combines computer

vision, analytics, a web dashboard, and automation workflows, performance and reliability matter as much as correctness. For this project, the main quality constraints are:

- **Performance:** maintain real-time behavior and low alert latency.
- **Scalability:** support multiple feeds and independent feed lifecycle control.
- **Reliability:** tolerate transient source failures and integration interruptions.
- **Usability:** provide clear web workflows for monitoring and configuration.
- **Security:** protect integration channels and service endpoints.
- **Maintainability:** keep modular code organization and testable interfaces.

These quality attributes are treated as engineering contracts rather than only design intentions, and the corresponding non-functional requirement quality matrix is summarized in Table 2.3.

ID	Quality	Target indicator	Validation
NFR-01	Performance	Keep the loop near real time and limit alert delay.	Runtime observation under representative feed scenarios.
NFR-02	Scalability	Support independent control for multiple feed contexts.	Feed registry behavior and state-transition checks.
NFR-03	Reliability	Recover gracefully from camera drops and webhook failures.	Reconnect paths and retry/fallback test scenarios.
NFR-04	Usability	Keep setup and monitoring clear for non-developer users.	Dashboard integration tests and workflow walkthroughs.
NFR-05	Security	Restrict integration access through secrets and credentials.	Webhook secret validation and credential checks.
NFR-06	Maintainability	Keep the architecture modular and testable.	Code review and test-suite coverage by module.

Table 2.3: Non-functional requirement quality matrix

2.3 Product Backlog

The product backlog was prioritized according to business value and technical dependency. To keep the implementation easy to follow, the items were grouped into three

domains that mirror the system architecture: backend processing, web supervision, and automation.

- **Backend domain:** perception pipeline, queue analytics, API and stream interfaces.
- **Web domain:** dashboard UX, live monitoring integration, zone editing workflows.
- **Automation domain:** n8n workflow integration, alert routing, webhook reliability.

The first sprints focused on backend foundations, because the dashboard and automation layers depend on a stable metric contract. This order reduced integration risk and made later validation clearer.

2.4 Global Use Case Diagram

The global use case diagram gives a high-level view of how the administrator and manager interact with the platform during configuration, monitoring, and alert handling, as shown in Figure 2.1.

Chapter 2 Use Case Diagram

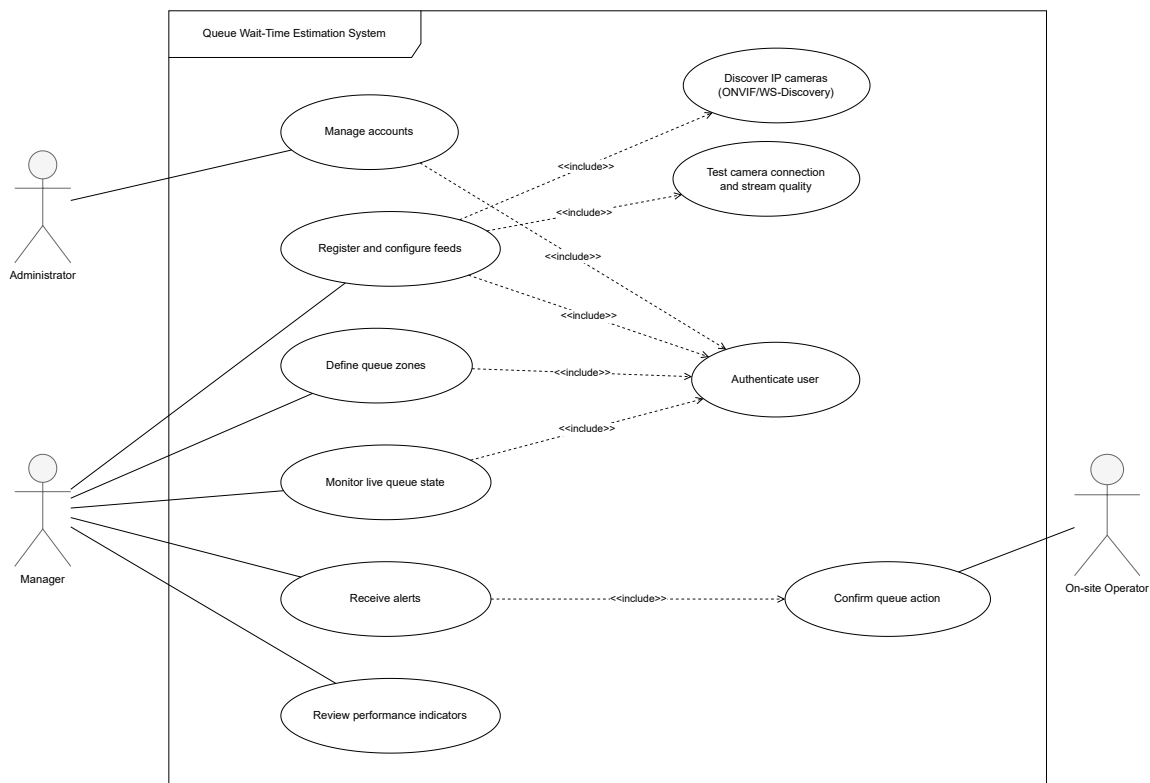


Figure 2.1: Global use case diagram of the queue wait-time estimation system.

At the system level, the use cases can be grouped into four interactions: configuration, live monitoring, analytics interpretation, and alert handling/resolution, with the main actor responsible for each use case listed in Table 2.4.

2.5 Sprint Scope Overview

We organized the project into four consolidated sprints so that the pipeline could be delivered step by step and checked at each stage. The sprint chapters that follow describe those implementation phases in detail, from perception and queue analytics to frontend integration and automation.

Use case ID	Use case summary	Primary actor
UC-01	Create, update, or remove manager accounts.	Administrator
UC-02	Register and configure a video feed source.	Manager
UC-03	Define or update queue zone geometry.	Manager
UC-04	Monitor live queue metrics and feed status.	Manager
UC-05	Receive and interpret warning or critical alerts.	Manager
UC-06	Dispatch operator after an alert.	Manager
UC-07	Confirm queue action completion.	On-site Operator (external)
UC-08	Review trends and performance indicators.	Manager

Table 2.4: Primary global use cases

2.6 Project Environment

2.6.1 Hardware and Software Environment

The hardware environment used during development is shown in Table 2.5, while the software and runtime layers are summarized in Table 2.6. The execution environment

Component	Type
Processor	Intel Core i5 12th Generation
Memory (RAM)	16 GB
Graphics card	NVIDIA GeForce RTX 4060
Storage	512 GB SSD

Table 2.5: Hardware environment

profile is detailed in Table 2.6.

Environment layer	Main tooling	Purpose
Backend runtime	Python ecosystem with vision and scientific libraries [8], [9], [10], [11], [12]	Perception, queue analytics, and integration dispatch.
Frontend runtime	Node/Bun-based TypeScript toolchain with Vite [13], [14], [15], [16]	Dashboard rendering, interaction workflows, and live-state synchronization.
Automation runtime	Self-hosted n8n with webhook endpoints [17]	Alert routing, cooldown logic, and external notifications.
Persistence/runtime support	SQLite database store (metrics, metadata, alerts) [18]	Historical analysis and system state continuity.

Table 2.6: Execution environment profile

2.6.2 Development Tools

The project also relied on a small set of development tools for editing, versioning, diagramming, container support, and report generation. Keeping them separate from the technology stack makes it easier to distinguish what was used to build the project from what the project actually runs on, as listed in Table 2.7.

Tool	Role in the project	Typical use
VS Code [19]	Main editor for the backend, frontend, and report sources.	Day-to-day implementation, review, and navigation.
Git / GitHub [20], [21]	Version control and change history tracking.	Commit management, branching, and collaboration.
draw.io [22]	Diagram authoring for architecture and UML models.	Use case, UML class, and system architecture diagrams.
Docker / Docker Compose [23]	Local service orchestration and reproducible setup.	Supporting services such as MediaMTX and n8n.

Table 2.7: Development tools used in the project

2.6.3 Technologies Used

The technology stack follows the main stages of the system, as summarized in Table 2.8.

Layer	Main technologies	Role in the project
Vision pipeline	YOLO26, ByteTrack, and OpenCV [2], [12], [24]	Detects people, tracks them across frames, and prepares the visual input for queue analysis.
Queue analytics	NumPy, Pandas, SciPy, and M/M/1 modeling [1], [9], [10], [11]	Computes arrival rate, service rate, and wait time.
Backend API	FastAPI and Pydantic v2 [25], [26]	Exposes typed endpoints, validates payloads, and coordinates the backend services.
Live transport	WebRTC, MediaMTX WHEP gateway, and MJPEG fallback [27], [28]	Delivers low-latency video streaming while keeping a simpler HTTP fallback available.
Frontend	React, TypeScript, Vite, and Tailwind CSS [15], [16], [29], [30]	Implements the dashboard, user interactions, and responsive presentation layer.
Automation	n8n workflows and Telegram notifications [17], [31]	Routes alerts and sends operational messages to users.
Persistence	SQLite [18]	Stores metrics, metadata, and alert history for later analysis.

Table 2.8: Technologies used in the project

This selection keeps the implementation modular from perception through automation.

2.7 Software Architecture

2.7.1 Overall Architecture

The architecture is described as four stages that match the flow of data through the system, from video acquisition to alert delivery, as illustrated in Figure 2.2.

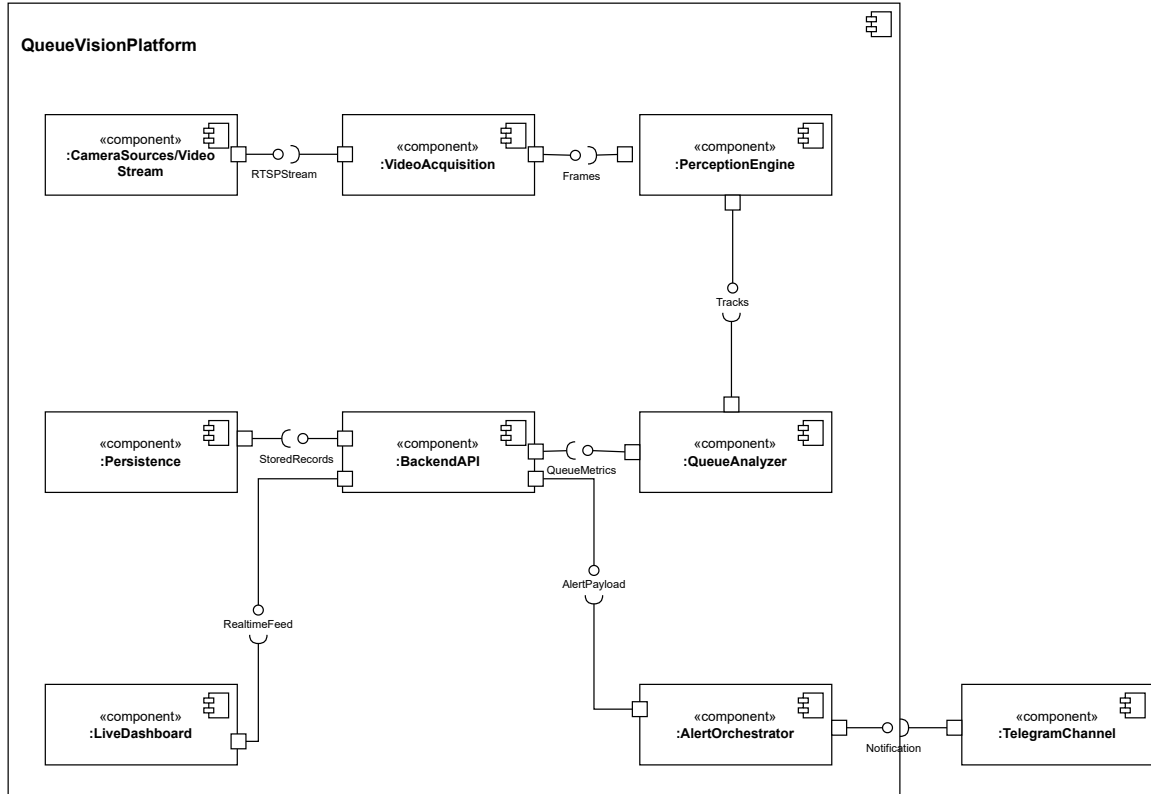


Figure 2.2: Overall system architecture from video acquisition to alert delivery.

1. **Video acquisition:** camera sources and video streams (RTSP, ONVIF, and local files) feed into the video acquisition component, which handles frame capture and stream management [32], [33].
2. **Perception:** the perception engine runs detection and tracking on each captured frame and exposes tracked identities through a defined interface to the queue analyzer.
3. **Analysis:** the queue analyzer receives tracked events and computes queue metrics (λ , μ , W) using adaptive wait-time estimation logic.
4. **Output and integration:** the backend API acts as the hub that receives queue metrics and distributes them across three outputs — the live dashboard via a real-time feed, the persistence layer (SQLite) for historical records, and the alert orchestrator (n8n) which routes alert payloads to the Telegram notification channel.

This separation keeps the four stages independently testable. Queue formulas in

the analyzer depend only on perception events; the backend API depends only on the metric contract; and the dashboard, persistence, and alert orchestration are each driven by the API without depending on each other.

2.7.2 Backend Architecture

The backend exposes REST and websocket interfaces for feed lifecycle control and live metrics. It includes modules for detection orchestration, tracking, queue analysis, threshold evaluation, webhook delivery, and camera-source management.

Feed-level orchestration lets each camera be started, stopped, or refreshed independently. If one feed fails, the others keep running, and the dashboard still receives a predictable event stream.

We treat persistence and integration as core responsibilities rather than add-ons: a durable backend state store preserves operational context, while webhook delivery turns queue events into external notifications.

2.7.3 Frontend Architecture

The frontend is built as a component-based interface with typed API integration, feed-oriented dashboard composition, and websocket-driven synchronization. Data access is isolated in hooks and client utilities, while pages and reusable components focus on presentation and interaction.

That separation keeps API changes localized to the integration layer and lets the visual components stay stable even when the backend contract changes. It also makes the dashboard easier to extend when new feed controls or indicators are added.

2.8 Conclusion

This chapter establishes the functional baseline, the quality constraints, and the architecture choices that guide the rest of the report. The next chapters turn these requirements into perception, connectivity, automation, and evaluation work.

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3.1 Introduction

Sprint 1 built the full path from raw video to actionable queue indicators. It started by detecting people, tracking them across frames, and filtering observations to the queue zone. Once that perception baseline was reliable, the sprint moved to queue interpretation: estimating how fast people arrive, how quickly service handles them, and how long a new arrival should expect to wait.

This combination was deliberate. Perception without analytics produces only a count; analytics without stable perception produces unreliable rates. By delivering both in one sprint, the project established a trustworthy metric pipeline that every later sprint could consume directly.

3.2 Sprint Objectives

This sprint was focused on two complementary groups of outcomes:

Perception objectives:

- detect people reliably in live and recorded video,
- preserve identity continuity across frames,
- filter detections to the configured queue zone,
- deliver a stable perception baseline for analytics.

Analytics objectives:

- estimate arrival and service rates from tracked events,
- compute wait time in a way that remains meaningful during changing conditions,
- protect the runtime from empty or unstable windows,
- publish a metric contract that later layers can consume directly.

These objectives define the core of the system. If the perception layer is noisy or the analytics layer is fragile, every later metric and operational decision becomes harder to trust.

3.3 Sprint Backlog

The Sprint 1 backlog separates manager-facing user stories from technical implementation tasks. User stories represent stakeholder needs and value; technical tasks are internal infrastructure concerns. This structure ensures that traceability focuses on business outcomes while documenting all delivery work.

3.3.1 User Stories

The user stories in Sprint 1 represent manager needs for reliable perception and analytics, as shown in Table 3.1.

User StoryID	User Story	TaskID	Task
US1.1	As a manager, I want to detect people in the video stream so that queue observations start from a reliable signal.	T1.1	Configure the detector to identify people in live and recorded video.
US1.2	As a manager, I want to keep the same person identifiable across frames so that counts remain consistent.	T1.2	Integrate tracking to preserve identity across consecutive frames.
US1.3	As a manager, I want to restrict counting to queue zones so that irrelevant movement does not affect the queue state.	T1.3	Define queue zones as polygons and filter detections by zone.
US1.6	As a manager, I want queue rates estimated from tracked people so that the dashboard can interpret flow intensity.	T1.6	Group tracked events into short windows and compute arrival and service rates.
US1.7	As a manager, I want wait time computed adaptively so that the result remains useful in real conditions.	T1.7	Apply the M/M/1 formula when arrivals are below service capacity and fallback to occupancy-based estimation otherwise.
US1.9	As a manager, I want a clear metric output so that the dashboard can reuse these values without reprocessing them.	T1.9	Publish rate estimates, wait time, and people-in-zone occupancy.

Table 3.1: Sprint 1 user stories

3.3.2 Technical Tasks

The technical tasks in Sprint 1 represent internal implementation concerns needed to support the user stories.

Task ID	Task	Subtask ID	Subtask
T1.4	Connect capture, detection, tracking, zone filtering, and display into one continuous flow.		Implement runtime orchestration that integrates all perception components.
T1.5	Expose runtime settings so that the pipeline can be adapted without code changes.		Provide configurable parameters for frame handling and visualization.
T1.8	Add guards for empty windows, low counts, and bounded fallback output to protect calculations from division-by-zero and noise.		Implement defensive bounds checking in the analytics layer.

Table 3.2: Sprint 1 technical tasks

3.4 Sprint Analysis

3.4.1 Runtime Scenarios

Sprint 1 is best understood as a runtime pipeline that converts each frame into queue metrics. The scenarios below describe the conditions that the activity and sequence diagrams have to cover.

Four key scenarios are addressed in Sprint 1:

1. **Person entry into queue zone:** when a person enters the queue area, the system treats that person as a possible arrival.
2. **Consistency across frames:** when the same person appears in consecutive frames, the system keeps the identification stable and avoids double counting.
3. **Zone filtering under mixed scenes:** people outside the queue area are ignored so the queue state stays focused on the relevant part of the scene.
4. **Queue behavior under varying load:** when arrivals stay below service, the system

uses the M/M/1 formula; when arrivals exceed service, the system switches to a conservative occupancy-based estimate without requiring an explicit stability output.

3.5 Design

3.5.1 Activity Diagram

The Sprint 1 activity flow from frame ingestion to queue metrics output is illustrated in Figure 3.1.

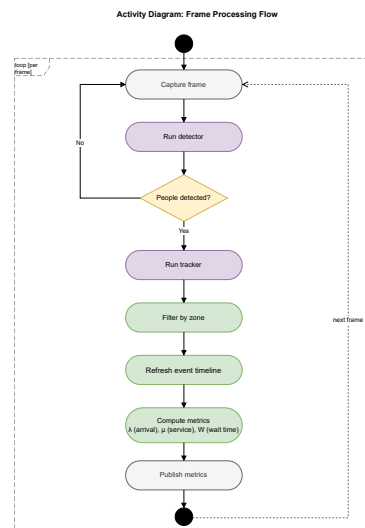


Figure 3.1: Sprint 1 activity flow from frame ingestion to queue metrics output.

The activity starts from a video source, whether it is a local file or a live camera stream. The system then detects people, keeps track of them, checks whether they are inside the queue area, and passes the filtered result to the queue analysis stage, which computes arrival rate, service rate, and expected wait time.

3.5.2 Analytics Activity Diagram

The detailed queue analytics logic is shown in Figure 3.2. This diagram focuses on the dynamic step that transforms tracked people inside the zone into the metric payload

consumed by the dashboard and later automation layers.

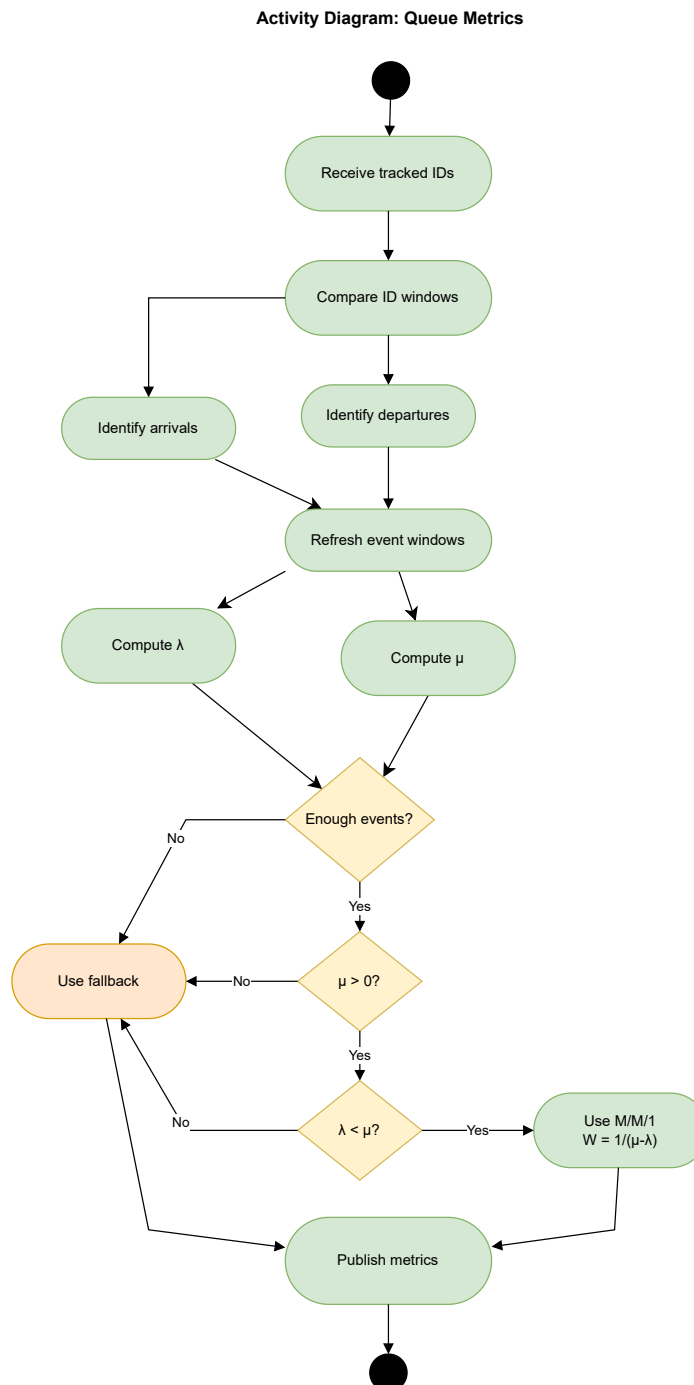


Figure 3.2: Sprint 1 analytics activity diagram for queue metric generation.

The analytics activity compares the current tracked identifiers with the previous observation window to identify arrivals and departures. These events are stored in short windows, then converted into λ and μ . Before applying the wait-time formula, the system checks whether enough events were observed, whether μ is positive, and

whether arrivals remain below service capacity. If these guards are satisfied, it uses the M/M/1 expression; otherwise, it falls back to a bounded occupancy-based estimate. This maps directly to US1.6, US1.7, T1.6, T1.7, and T1.8.

3.5.3 Sequence Diagram

The Sprint 1 sequence from video ingestion to queue metric publication is shown in Figure 3.3.

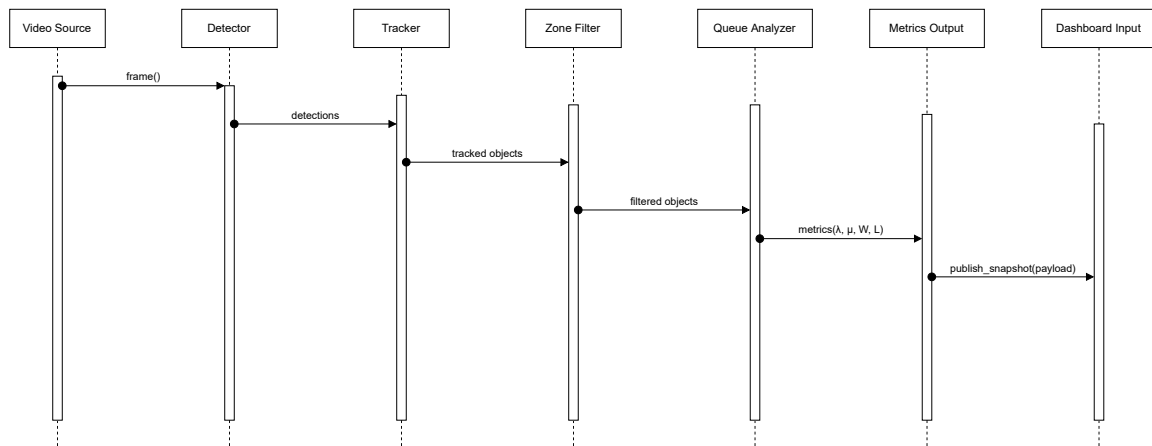


Figure 3.3: Sprint 1 sequence diagram from video ingestion to queue metric publication.

In practice, each frame moves through source handling, detection, tracking, zone checking, rate estimation, and metric publication. That ordered flow is what lets Sprint 1 deliver both a perception baseline and an analytical output in one pass.

3.5.4 Operational Scenarios

The queue analysis is easiest to understand through the situations it must handle every time a new metric window is evaluated:

1. **Normal queue:** arrivals stay below service, so the M/M/1 formula gives a precise estimate based on measured rates.
2. **Growing backlog:** arrivals start to approach service, so the system seamlessly switches to a conservative estimate based on current occupancy.

3. **Empty or stalled queue:** no service events are available, so the output stays bounded and downstream logic receives safe, interpretable values.

These scenarios explain why the sprint computes wait time with adaptive logic: a single metric that responds correctly to real-world queue conditions without requiring separate state signals.

3.6 Implementation

3.6.1 Detection Layer

The detection layer focuses on finding people in each frame while keeping the process adjustable for different machines. It is tuned to balance speed and accuracy, so the system stays responsive without losing the main detections that matter for the queue.

3.6.2 Tracking Layer

The tracking layer keeps the same person identifiable from one frame to the next. This is what makes arrival and departure counts more reliable than simply reading detections in a single frame [24].

3.6.3 Zone-Aware Filtering

Zone logic defines the queue area as a polygon on the frame and keeps only the people inside it. This avoids counting unrelated movement elsewhere in the scene and makes the measurement more relevant to the waiting line.

3.6.4 Main Loop Integration

The pipeline integration is handled in the runtime orchestration layer. The runtime supports:

- handling local files, webcams, and live camera streams,
- adjusting how often frames are processed so the system stays responsive,
- showing detections, IDs, the queue zone, and metrics on screen,
- running without a visible interface when needed for later integration.

3.6.5 Queue Event Windows and Rate Estimation

The queue-analysis step uses short time windows to count arrivals and departures. Let N_a and N_d be the number of events observed in windows T_a and T_d :

$$\lambda = \frac{N_a}{T_a}, \quad \mu = \frac{N_d}{T_d}$$

From these counts, the system estimates the arrival rate and the service rate. To avoid unstable values at the beginning of a run, the implementation waits until enough events have been observed before returning non-zero rates.

3.6.6 Wait-Time Logic and Adaptive Estimation

The system estimates wait time using an adaptive formula that responds to queue conditions:

$$W = \begin{cases} \frac{1}{\mu - \lambda} & \text{if } \lambda < \mu \text{ and } \mu > 0 \\ \frac{L}{\mu} & \text{otherwise} \end{cases}$$

where L is the current number of people in the zone. We use the M/M/1 sojourn-time expression because the project models each monitored lane as a single service channel with one effective queue, which matches the cashier-style setting studied in the report. This formula is also easy to interpret during a defense: it depends directly on the measured arrival and service rates, it gives a closed-form estimate without needing a large historical dataset, and it stays aligned with the real-time pipeline

used by the tracker. Other models were less appropriate here: M/M/c would assume multiple parallel servers, which is not the default lane setup, and Little's Law is useful for validation but does not directly produce a forward wait-time estimate from the instantaneous rates alone.

The adaptive formula ensures the output always remains safe and interpretable: when arrivals are below service capacity, the formula is precise; when arrivals exceed capacity, it gracefully degrades to a conservative estimate based on current queue size. When no service events are observed, the implementation returns zero so that downstream components do not use an unreliable value.

3.6.7 Runtime Metric Contract

The final metric output includes:

- point estimates (λ, μ, W) ,
- people-in-zone occupancy.

These values are then reused by logs, overlays, webhook payloads, and dashboard events [1], [12].

3.7 Conclusion

Sprint 1 delivered the full path from raw video to operational queue indicators. People can be detected, followed across frames, counted inside the queue area, and the resulting events are turned into arrival rates, service rates, and wait-time estimates. That gives the next sprint a trustworthy analytical contract to build on for camera integration and API services.

3.8 Sprint Retrospective

The main achievement of this sprint was connecting perception to interpretation in one delivery. The tracker and zone logic reduce false continuity problems, while the queue model provides an adaptive wait-time formula that remains interpretable across all operating conditions. A mathematically correct wait-time result is not enough if the input window is empty or the queue is overloaded, so the implementation had to treat safety and interpretability as equal concerns alongside detection accuracy.

The value of this sprint is that it gives the rest of the system a trustworthy metric layer. Once the queue state is expressed through rates and adaptive wait-time estimation, the dashboard and automation work can operate on structured signals instead of raw counts.

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4.1 Introduction

Sprint 2 expanded the project from a single local pipeline into a platform that can work with several camera sources. At this stage, the backend became responsible for camera onboarding, feed control, and live delivery, so the rest of the application could communicate with it in a more structured way.

4.2 Sprint Objectives

This sprint had four main delivery goals:

- make external camera sources usable inside the project,
- expose feed management through a predictable API contract,
- keep live delivery available through a fallback-aware transport strategy,
- preserve feed state and recovery behavior across restarts.

These goals are important because this sprint turns the backend into a reusable service layer rather than a one-off processing script.

4.3 Sprint Backlog

The Sprint 2 backlog separates manager-facing user stories from technical implementation tasks. User stories represent stakeholder needs and value; technical tasks are internal infrastructure concerns.

4.3.1 User Stories

The user stories in Sprint 2 represent manager needs for secure and automated camera integration, as shown in Table 4.1.

User StoryID	User Story	TaskID	Task
US2.1	As a manager, I want IP camera streams connected securely so that live sources can be reused reliably.	T2.1	Add authentication, transport control, reconnection, and snapshots for RTSP feeds.
US2.2	As a manager, I want cameras discovered automatically so that stream addresses do not need to be hardcoded.	T2.2	Detect IP cameras and resolve their stream addresses.
US2.3	As a manager, I want feed management exposed through the API so that camera lifecycle actions are centralized.	T2.3	Expose feed control, metadata handling, and diagnostic operations.
US2.4	As a manager, I want live delivery to stay available so that the dashboard can keep showing the active feed.	T2.4	Support live delivery through real-time events and a layered video transport strategy.

Table 4.1: Sprint 2 user stories

4.4 Design

4.4.1 Sequence Diagrams

The Sprint 2 sequence for feed orchestration and live event distribution is illustrated in Figure 4.1.

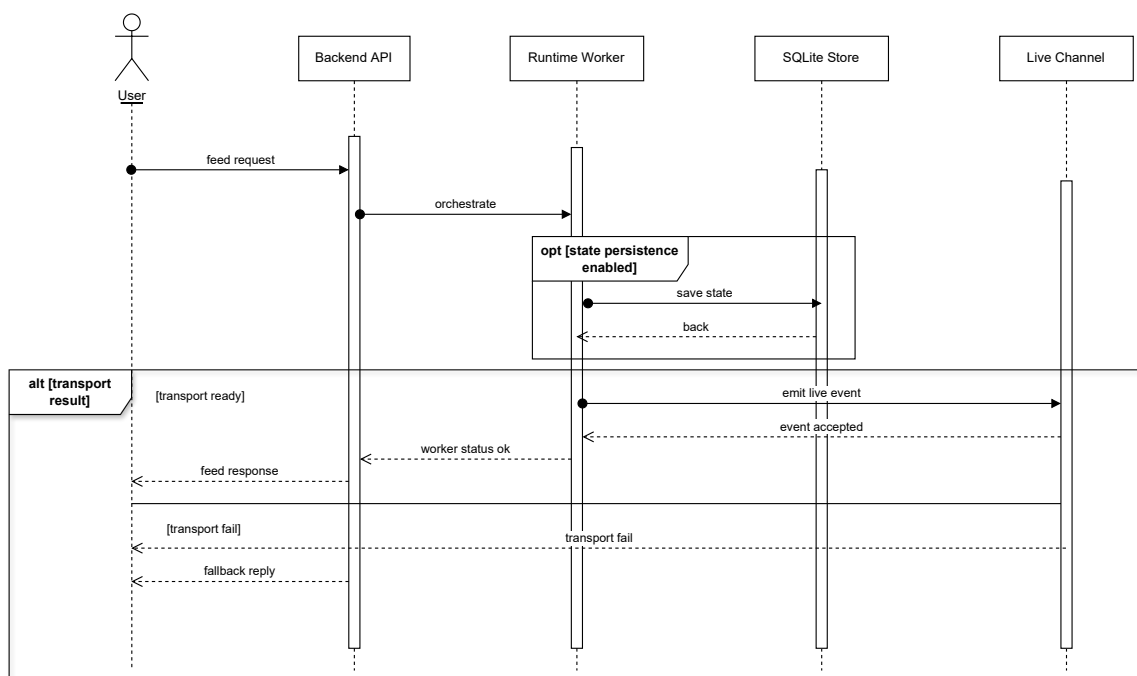


Figure 4.1: Sprint 2 sequence diagram for feed orchestration and live event distribution.

The running sequence is straightforward: a frontend action reaches the backend, the

feed state is updated, the worker process changes accordingly, and the new status is sent back to the live interface.

4.4.2 Backend Class Diagram

The Sprint 2 backend class diagram used by feed orchestration and persistence is shown in Figure 4.2.

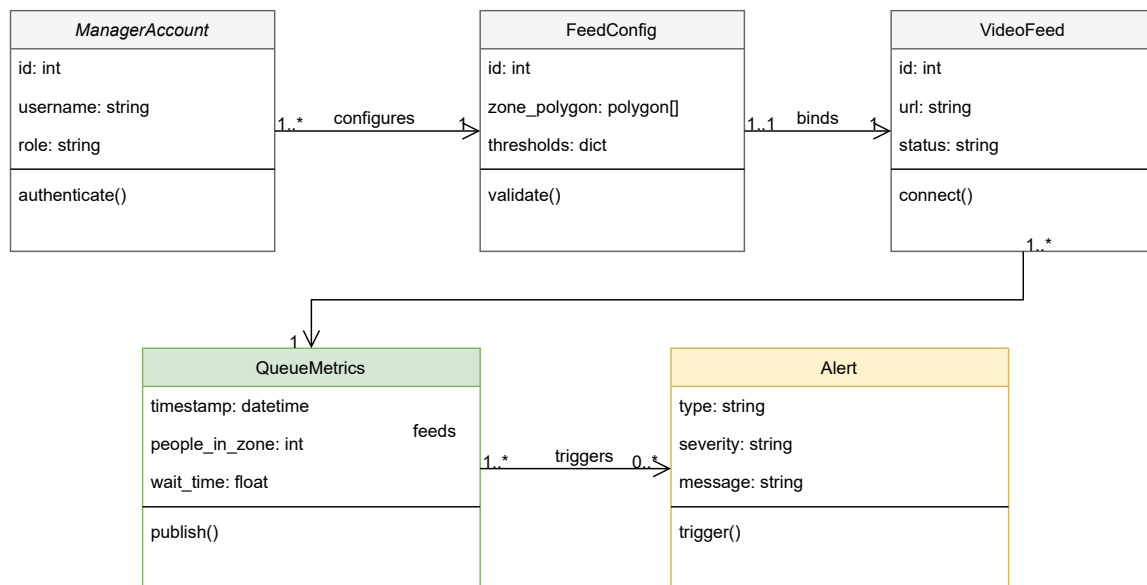


Figure 4.2: Sprint 2 backend class diagram showing the core entities used by feed orchestration and persistence.

This structural view complements the sequence diagram by showing how the back-end keeps manager accounts, feed configurations, video feeds, queue metrics, and alerts in a cohesive domain model. It is especially relevant in Sprint 2 because the chapter introduces authenticated API access, SQLite persistence, and feed-state recovery.

4.5 Implementation

4.5.1 Camera Source Integration

This part of the sprint focuses on making external cameras usable inside the project. It supports:

- handling camera credentials without exposing them in logs,
- choosing between TCP and UDP transport depending on the situation,
- checking the connection and capturing snapshots,
- reconnecting automatically when the stream is interrupted.

The source-handling layer routes local files and webcams through one path and live camera streams through another, which keeps the source logic easier to maintain.

4.5.2 Authenticated API Access

Sprint 2 also introduced session-aware access control in the backend API. Managers can only interact with the feeds that belong to their account, while administrators keep the broader management scope needed for account governance. The same identity context is applied to feed listing, feed creation, feed lifecycle actions, stream snapshots, and websocket subscriptions so the runtime respects the ownership rules defined earlier in the report.

4.5.3 Camera Discovery and Stream Setup

The camera discovery step is used to find available cameras automatically and recover their stream information. This avoids hardcoding stream addresses and makes onboarding easier when new IP cameras are added. In the implementation, this is not a generic object-detection step; it is a network-source discovery workflow based on ONVIF and RTSP. The backend sends WS-Discovery probes on the local network, extracts device metadata from ONVIF probe responses, resolves the advertised media service, requests RTSP stream URIs through ONVIF media SOAP calls, and then validates the selected stream with OpenCV before it is registered as a feed [12], [32], [33].

In the practical setup, the workstation and the IP camera were connected to the same local network through a router, which made the device reachable for discovery

and RTSP validation. The test source used during this stage was a Digital Watchdog IP camera, allowing the connectivity workflow to be validated against a real camera feed.

The resulting onboarding logic is summarized in Figure 4.3.

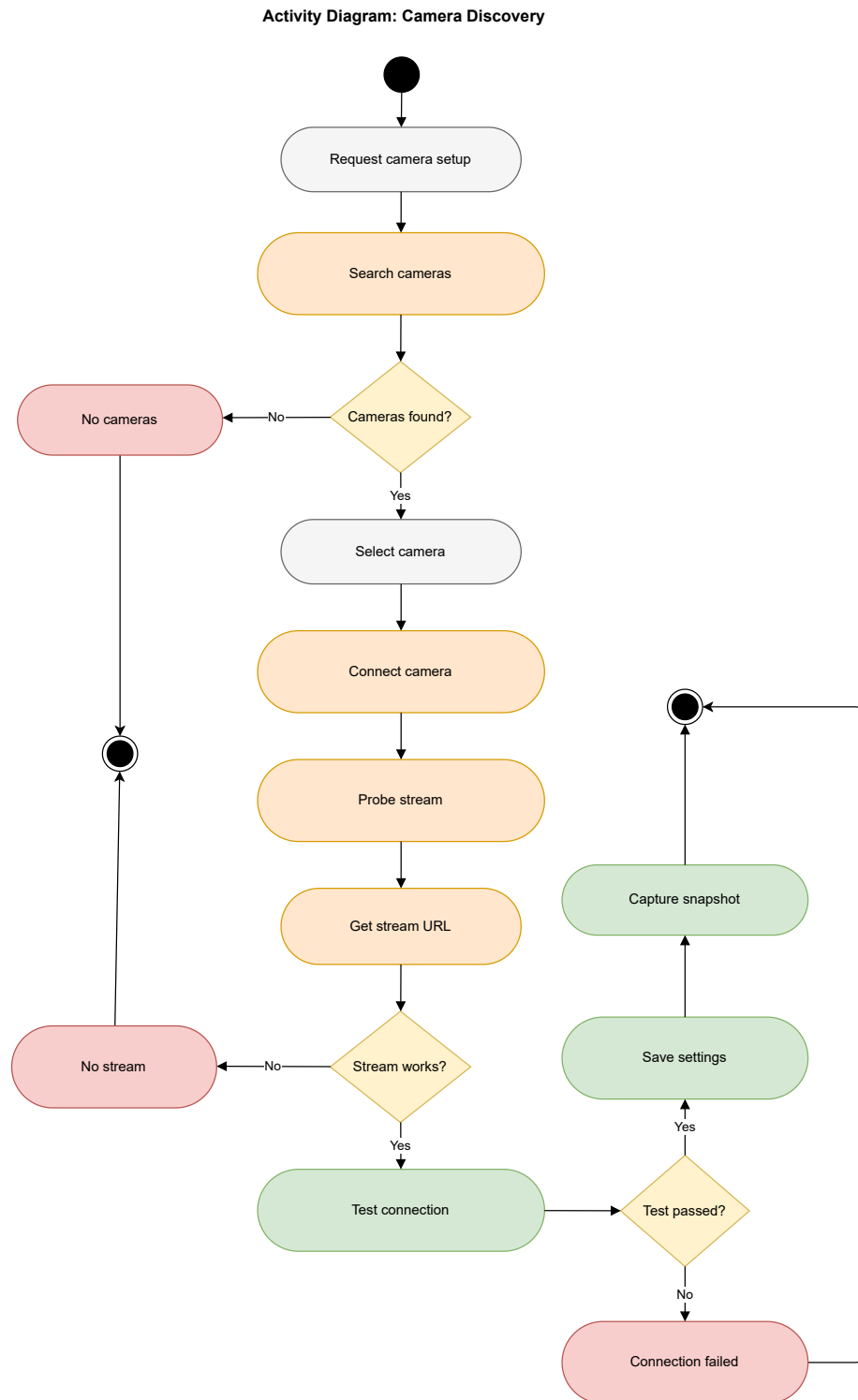


Figure 4.3: Sprint 2 camera discovery activity diagram for ONVIF discovery, RTSP resolution, and stream validation.

The same flow is exposed through backend endpoints for ONVIF discovery, stream resolution, camera testing, and RTSP snapshots. Credentials are accepted as runtime inputs and are kept out of user-facing logs, while the RTSP transport option allows TCP or UDP to be selected depending on stream stability and latency needs.

4.5.4 API Layer and Data Contract

The API layer acts as the bridge between the backend services and the rest of the project. It provides structured inputs and outputs so the frontend can interact with the backend in a predictable way. The main groups of operations are:

- feed management, including creation, listing, start, stop, restart, and deletion,
- source onboarding, including RTSP checks, snapshots, and camera discovery,
- metadata management for establishments and cashiers,
- integration and health checks,
- live transport support for real-time viewing.

4.5.5 Live Video Delivery

To reduce viewing latency while still keeping a fallback path, Sprint 2 introduces a transport system that chooses the most suitable stream mode for each feed [27], [34]. The backend tells the frontend whether live viewing is ready, whether WebRTC is available, and which fallback should be used when the higher-quality path cannot start. When WebRTC is available, the backend forwards the browser offer to the media server and returns the response needed to start the stream [28]. If the feed does not exist, is not running, or the media service cannot answer, the backend reports the condition clearly so the frontend can switch to the fallback view instead of failing silently. The configuration stays flexible through environment settings such as the media URL, control URL, timeout, and feature flag.

4.5.6 Runtime Control and Live Events

Runtime control sits inside the API orchestration layer. It keeps track of feed state, synchronizes persistence, manages the worker lifecycle, and dispatches events to the rest of the system. The live channel publishes updates such as snapshots, feed status changes, metric updates, alert notifications, and system warnings.

For the video surface, the system keeps a shared fallback stream so that viewing does not break when the runtime mode changes.

4.5.7 Backend Persistence Architecture

The backend persistence architecture is implemented in SQLite, with a schema that preserves establishments, checkout counters, feed configurations, video sessions, alerts, users, refresh sessions, and auth audit records. This storage design supports both operational recovery and historical analysis, while the runtime layer uses it to restore stale sessions and preserve manager metadata across restarts.

4.5.8 Persistence and Recovery

The persistence layer stores feed configuration, metadata, video sessions, and archived alerts in SQLite. When the backend starts, it reloads saved feeds and safely resets any stale running sessions so the system begins from a consistent state.

4.6 Conclusion

By the end of Sprint 2, the backend could onboard cameras, manage feed state, and send live updates to the frontend without depending on hardcoded source details. That made the web UI work in Sprint 3 much easier to integrate.

4.7 Sprint Retrospective

This sprint clarified the role of the backend as the integration hub. It now handles camera identity, live transport, recovery, metadata, and state synchronization in one place, which reduces coupling in the rest of the system.

The main lesson is that connectivity is not only about reaching the camera. It is also about preserving a usable application state when streams fail, restart, or need to switch transport modes.

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5.1 Introduction

Sprint 3 covered the frontend work from the first design decisions and page structure to the live backend integration that made the dashboard operational. The sprint started by defining the visual direction and navigation model, then connected the interface to backend services and runtime events so the dashboard could react to live data, transport changes, and feed lifecycle actions.

This two-phase approach was intentional. Settling the design system and page skeleton first meant that the live integration work could focus on behavior and reliability instead of redesigning the interface at the same time.

5.2 Sprint Objectives

This sprint was organized around six outcomes spanning design and live integration:

Design and scaffolding objectives:

- define a coherent visual language for the dashboard,
- prepare the main page layouts before live data integration,
- build reusable UI structures that can support later features.

Live integration objectives:

- make the dashboard consume the backend contract without ad hoc transformations,
- keep the interface responsive when live transport is available, delayed, or interrupted,
- validate that monitoring, feed management, and fallback behavior remain understandable for the operator.

5.3 Sprint Backlog

The Sprint 3 backlog separates manager-facing user stories from technical implementation tasks. User stories represent stakeholder needs and value; technical tasks are internal infrastructure concerns.

5.3.1 User Stories

The user stories in Sprint 3 represent manager and admin needs for dashboard interfaces and live monitoring, as shown in Table 5.1.

User StoryID	User Story	TaskID	Task
US3.1	As a manager, I want a consistent visual style so that the dashboard feels like one coherent product.	T3.1	Define colors, typography, spacing, and reusable component rules.
US3.2	As a manager, I want the main page layouts prepared so that onboarding, monitoring, editing, and analytics screens are available.	T3.2	Prepare the landing page, dashboard, setup wizard, zone editor, analytics, and settings layouts.
US3.3	As an admin, I want to create and maintain manager accounts so that system access is centrally controlled.	T3.3	Build the admin dashboard for manager account creation, editing, and deletion.
US3.4	As a manager, I want the dashboard connected to backend responses so that the interface reflects live operational data.	T3.4	Implement the API integration layer and typed request handling.
US3.5	As a manager, I want to manage feeds from the interface so that feed lifecycle actions stay centralized.	T3.5	Connect login state, protected routes, and feed control actions to the dashboard.
US3.6	As a manager, I want live updates without refreshing so that changes are visible immediately.	T3.6	Add websocket-driven live state updates and dashboard orchestration.
US3.7	As a manager, I want a suitable video transport chosen automatically so that the feed remains usable when conditions change.	T3.7	Add transport selection, fallback streaming, and reconnection logic.
US3.8	As a manager, I want the dashboard to stay usable during interruptions so that monitoring does not stop during transient failures.	T3.8	Add fallback polling and stale-data guards.

Table 5.1: Sprint 3 user stories

5.3.2 Technical Tasks

The technical tasks in Sprint 3 represent internal implementation concerns needed to support the user stories.

Task ID	Task	Subtask ID	Subtask
T3.10	Set up the frontend project with shared components and page scaffolding so that the interface can grow without duplication.		Create reusable UI component library and establish project structure.
T3.11	Validate dashboard setup and onboarding flows with integration tests to ensure the integration behaves correctly end to end.		Implement scenario-based tests for dashboard lifecycle and error paths.

Table 5.2: Sprint 3 technical tasks

5.4 Design

5.4.1 UI and UX Design

The project design specification introduces a “Terminal Minimal” style and a seven-page navigation model. The goal was to give the dashboard a simple visual identity that fits a monitoring application. The design language emphasizes:

- dark operational surfaces optimized for continuous monitoring,
- strong semantic colors for status and alert levels,
- card-based information density with consistent spacing and hierarchy,
- explicit workflow guidance in setup and zone-editing screens.

The designed page set includes the landing page, dashboard, setup wizard, zone editor, analytics, settings, and heatmap views. These screens helped define what the frontend had to support before live data arrived.

5.4.2 Sequence Diagrams

The Sprint 3 sequence for dashboard updates and alert surfacing is illustrated in Figure 5.1.

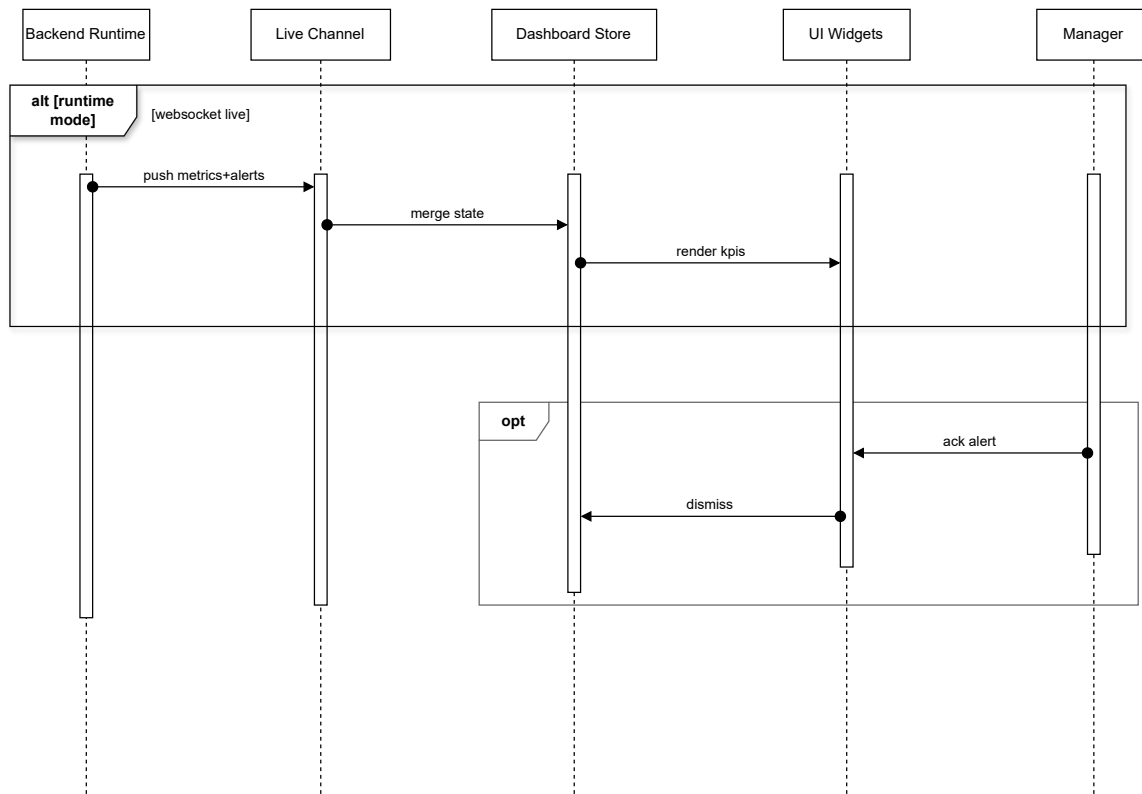


Figure 5.1: Sprint 3 sequence diagram for dashboard updates and alert surfacing.

Three main flows were implemented:

1. **Feed lifecycle:** a dashboard action updates the feed state and the new status is sent back to the interface.
2. **Live metrics:** metric updates are received from the backend and reflected immediately on the dashboard.
3. **Alert surfacing:** alerts reach the interface and appear in the activity and attention areas.

5.4.3 Dashboard Data-Flow Sequence Diagram

Figure 5.2 details how data moves between the user, the React dashboard, the API layer, the backend, the runtime worker, the websocket channel, and the media transport layer.

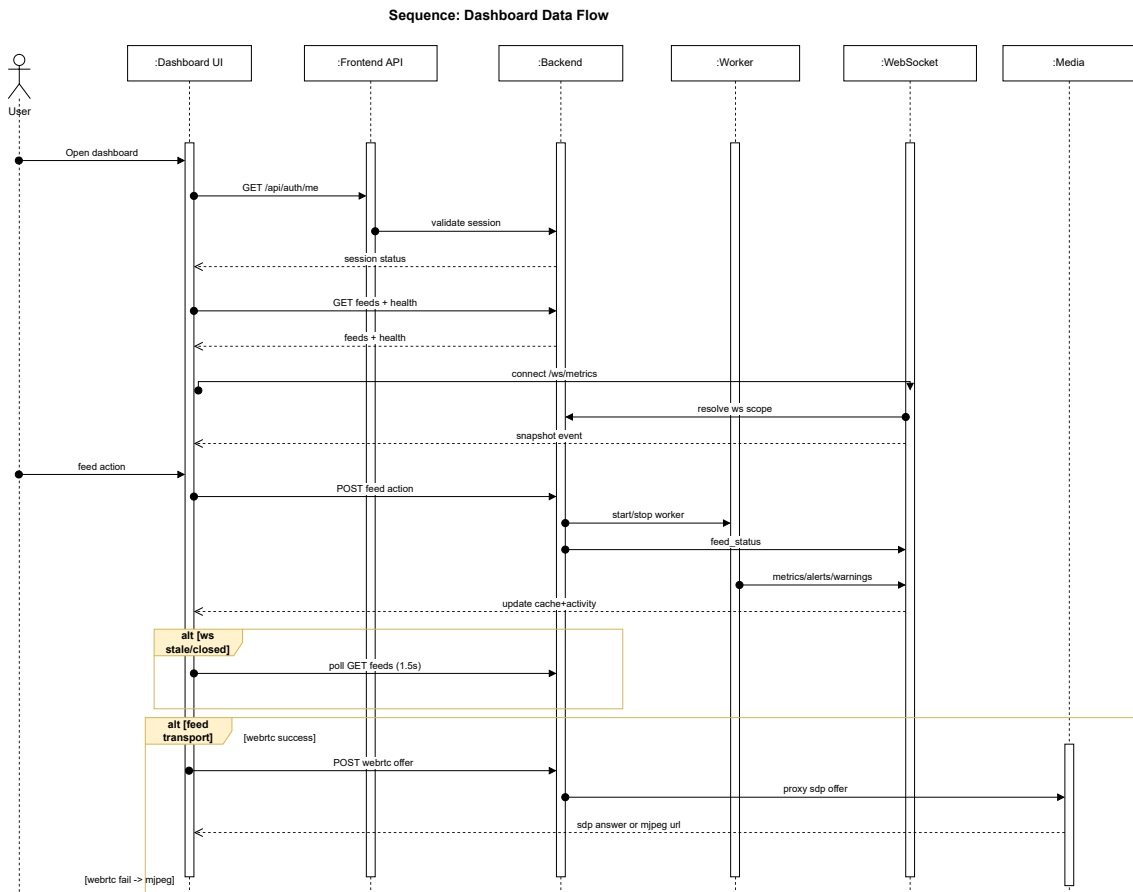


Figure 5.2: Sprint 3 detailed sequence diagram for dashboard data flow, live events, fallback polling, and video transport.

This sequence starts with protected dashboard access and initial REST queries for feeds and system health. The dashboard then opens the metrics websocket, receives an initial snapshot, and keeps its local state synchronized from feed lifecycle, metric, alert, and warning events. Feed actions are sent as REST commands and reflected back through websocket status events. When live events become stale, the frontend polls the feed list as a fallback. For video playback, the dashboard negotiates WebRTC through the backend when the transport is ready and otherwise uses the MJPEG stream endpoint as the simpler fallback.

5.4.4 Live Dashboard Activity Diagram

The frontend runtime behavior is summarized in Figure 5.3. This diagram complements the sequence view by showing how the dashboard decides between real-time websocket updates, fallback polling, feed actions, and video transport modes.

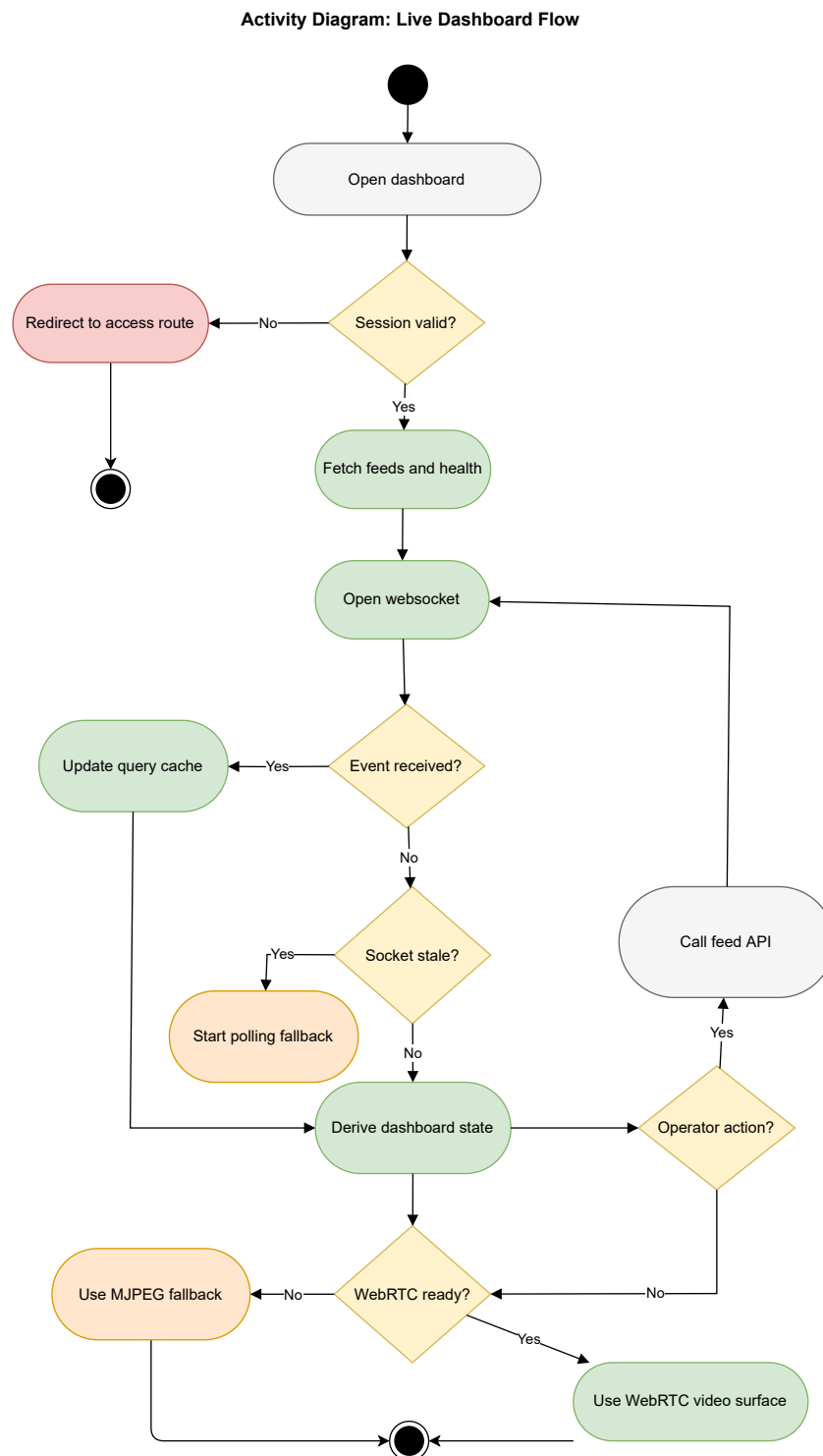


Figure 5.3: Sprint 3 live dashboard activity diagram for websocket updates, fallback polling, feed actions, and transport selection.

The activity begins with authenticated dashboard access, then loads the feed list and system health. After that, the websocket channel keeps feed status, metric updates, alerts, and warnings synchronized with the local query cache. If the websocket becomes

stale or unavailable while feeds are running, the dashboard temporarily polls the feed endpoint so the monitoring surface remains usable. In parallel, each feed card selects the best available video surface: WebRTC when the backend reports a ready stream, MJPEG when WebRTC is unavailable or fails, and a stored preview when the feed is not yet running.

5.4.5 Design Rationale

The design phase was kept as a foundation step rather than a full UI release. The goal was to settle navigation, component boundaries, and interaction patterns before live data and transport logic arrived. A clean shell was a useful deliverable because the integration work depended on it.

5.5 Implementation

5.5.1 Project Setup and Layout Skeleton

The implementation started by turning the design specification into an application skeleton. The workspace setup established the application entry point, protected navigation, the dashboard layout, and the shared components that all pages reuse.

The order of the pages was not arbitrary. The layout had to support a workflow that begins with onboarding, continues through configuration, and ends with live supervision. That is why the page skeleton was treated as an operational path rather than a list of unrelated screens.

5.5.2 Styling and Reusable Interface Rules

The visual system was defined once and then applied consistently across the dashboard. Spacing, typography, surface treatment, and semantic status colors were treated as

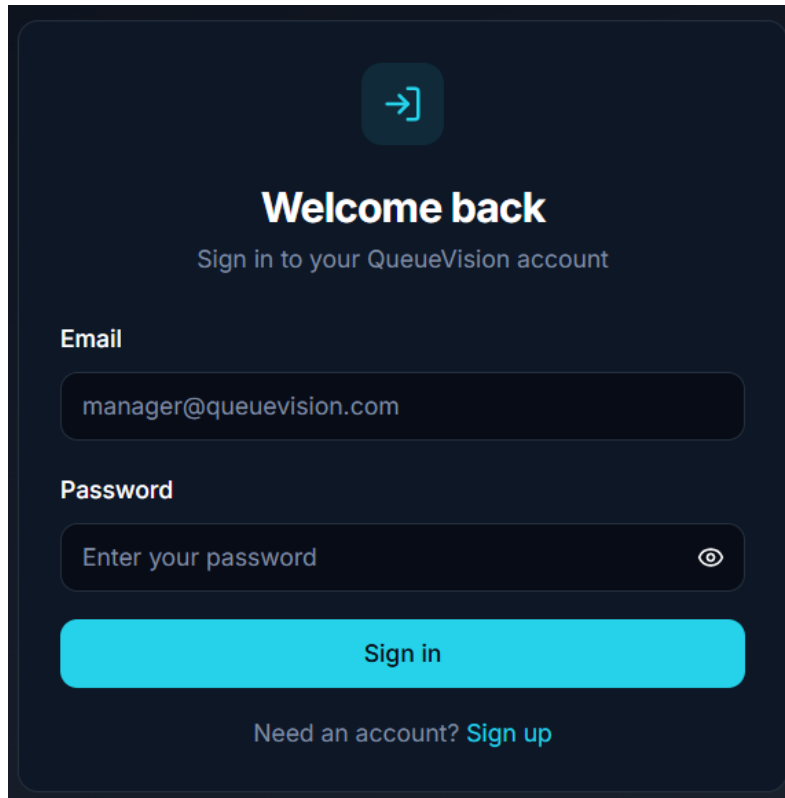
reusable rules rather than per-page decisions. This reduced design drift: when a status card, a table, or a setup control appears in several places, it behaves like the same product element instead of being restyled each time.

5.5.3 API Integration

The frontend relies on a dedicated API layer that keeps the communication with the backend structured. It defines the main data types used by the dashboard and centralizes request handling so errors can be managed consistently. This layer acts as the contract boundary: it prevents the UI from depending directly on backend implementation details and makes it easier to reason about feed lists, dashboard summaries, and management actions as explicit typed responses.

5.5.4 Authentication and Role-Based Navigation

Sprint 3 connected the login and sign-up flows to the dashboard. The auth provider clears cached data when the session changes, protected routes redirect users according to their role, and administrators are taken to the manager administration page while managers are routed to the monitoring dashboard. This keeps the frontend aligned with the backend access rules and avoids exposing the wrong interface to the wrong user, with the manager sign-in screen shown in Figure 5.4.



The image shows a dark-themed sign-in interface. At the top center is a teal icon of a right arrow pointing into a bracket. Below this is the heading "Welcome back" in white, followed by the subtitle "Sign in to your QueueVision account" in a smaller white font. The form contains two input fields: "Email" with the value "manager@queuevision.com" and "Password" with the placeholder "Enter your password". A teal "Sign in" button is positioned below the password field. At the bottom, there is a link that says "Need an account? Sign up" in white text.

Figure 5.4: Manager sign-in screen used to access the dashboard in Sprint 3.

The complementary manager account creation screen introduced in the same authentication flow is presented in Figure 5.5.

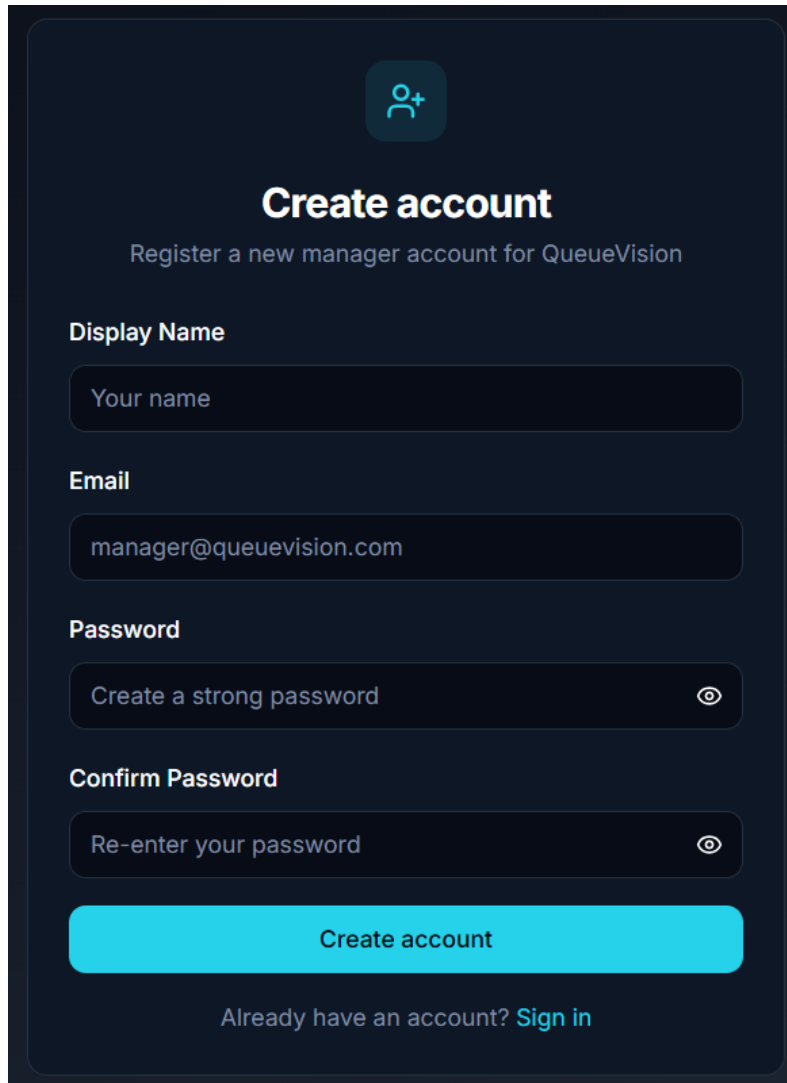
The image shows a 'Create account' form for QueueVision. At the top is a dark blue header with a white user icon and the title 'Create account'. Below the title is a subtitle: 'Register a new manager account for QueueVision'. The form contains four input fields: 'Display Name' with placeholder text 'Your name', 'Email' with placeholder text 'manager@queuevision.com', 'Password' with placeholder text 'Create a strong password' and an eye icon, and 'Confirm Password' with placeholder text 'Re-enter your password' and an eye icon. A large blue button labeled 'Create account' is positioned below the password fields. At the bottom, there is a link: 'Already have an account? Sign in'.

Figure 5.5: Manager account creation screen introduced as part of the Sprint 3 authentication flow.

5.5.5 Dashboard State Orchestration

A dedicated dashboard state layer coordinates data fetching and updates. From the received feed data, the dashboard derives indicators such as online feeds, total people, average wait time, and attention counts, then exposes them to the widgets.

The state layer also helps absorb transient data gaps. When a websocket event arrives late, the dashboard can keep displaying the last valid snapshot instead of dropping the screen into an empty state. That design choice improves usability during unstable network periods and keeps the monitoring surface trustworthy.

5.5.6 Live Update Reliability

The live-update layer includes:

- reconnection delays,
- a heartbeat check,
- a guard against stale data,
- fallback polling when live updates fail,
- grouped updates to keep the interface responsive.

When the websocket connection drops, the dashboard falls back to periodic polling instead of freezing, so operators still have a usable view during transient issues.

This fallback behavior was treated as a first-class feature rather than an error path. In a queue-monitoring context, the operator needs continuity more than perfect transport purity, so the UI must remain informative even when the preferred live path is temporarily unavailable.

5.5.7 Live Video Delivery

Live feed cards use a transport decision layer to choose the most suitable viewing mode. For each running feed, the frontend checks whether live viewing is ready. If the higher-quality live stream is available, it is used first; otherwise the dashboard falls back to the simpler video stream.

The live-streaming flow is handled through the browser's live video connection support [27]:

- create a connection for receiving video,
- gather the network information needed for signaling,
- send the browser offer to the backend,

- apply the backend response and switch the feed card to live state.

Reconnection uses increasing delays between attempts. When the problem is not recoverable, the system stops retrying and switches to the fallback stream so the dashboard remains usable.

When server-side annotations are active, the dashboard hides local overlays to avoid drawing the same boxes twice on the same video.

5.5.8 Dashboard UI Integration

The dashboard page brings together live feed controls, setup dialogs, source onboarding, KPI visualizations, and charts. The onboarding workflow supports both uploaded videos and ONVIF camera sources. For camera sources, the manager discovers devices, applies per-camera credentials when needed, resolves RTSP streams, tests the selected camera, captures a snapshot for zone tracing, and then stages the feed for batch launch. The emphasis was on making those elements work together as one operational interface.

The practical result is that a manager can move from onboarding to monitoring without leaving the dashboard context. That continuity mirrors the workflow described in the earlier requirements chapter, while Figure 5.6 shows the administrator screen used for creating and maintaining manager accounts.

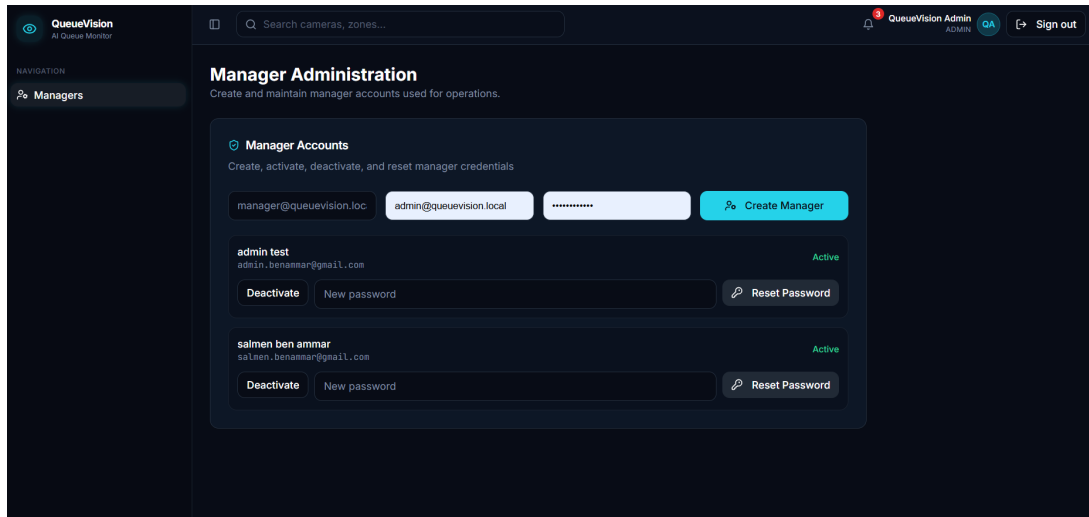


Figure 5.6: Administrator manager-administration screen for creating and maintaining manager accounts.

The manager-facing monitoring dashboard, including the live surface, feed summary cards, and recent activity panel, is presented in Figure 5.7.

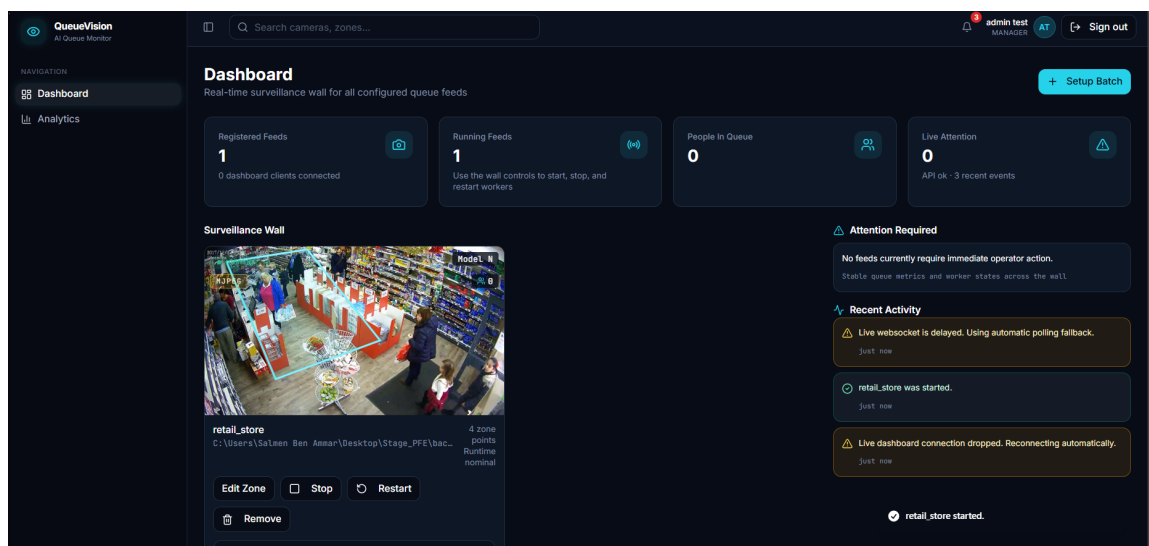


Figure 5.7: Manager dashboard showing the live monitoring surface, feed summary cards, and recent activity panel.

The onboarding workflow starts with the feed setup screen shown in Figure 5.8, where the manager prepares an uploaded video or an ONVIF camera source before zone tracing and model selection.

The screenshot shows a web interface titled "Stage Sources" with a close button (X) in the top right corner. Below the title is a descriptive text: "Enter the feed information for one source, optionally attach metadata, then continue through zone tracing, model selection, and review." A light blue box contains the instruction: "One source is prepared at a time. Finish this source, then continue to zone tracing and model selection."

The "Feed name" section has a text input field containing "Checkout 1".

The "Source type" section features two buttons: "Local MP4" (which is highlighted with a blue border) and "ONVIF Discovery".

Below the source type buttons is a section for uploading a local video. It includes a video camera icon, the text "Upload a local video", and a subtext: "Supported formats: MP4, MOV, AVI, and MKV. The backend stores the file and uses its path automatically." There is a button labeled "Choose Video File" and a status message "No video selected yet". A note below states: "Local videos are configured one at a time so each feed keeps a single zone and model."

A step-by-step guide is provided: "Step 1 uploads the source file into the backend workflow. Step 2 lets you draw the queue polygon on the video frame. Step 3 selects the model size."

A dashed rectangular box contains the text: "Pick one local video to see it listed here."

At the bottom right, there are two buttons: "Cancel" and "Continue to Zone".

Figure 5.8: Feed setup screen used to prepare an uploaded video or ONVIF camera source before zone tracing and model selection.

The queue polygon for the selected feed is defined through the zone tracing screen shown in Figure 5.9.

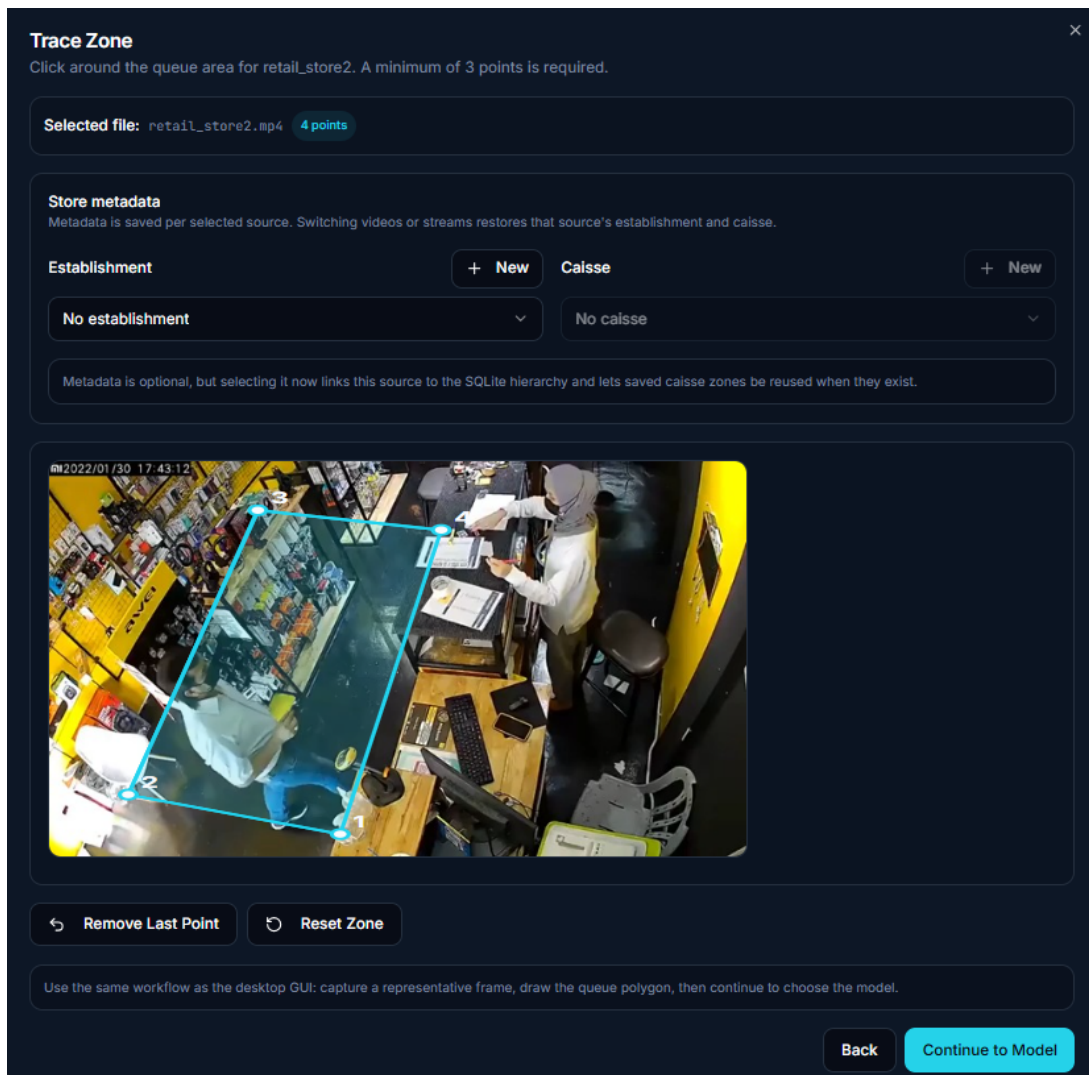


Figure 5.9: Zone tracing screen used to define the queue polygon for the selected feed.

Before launching the feed, the manager selects the YOLO model size through the screen shown in Figure 5.10.

Choose Model

×

Finalize retail_store2 with the YOLO model size you want to run.

SOURCE TYPE	FEED NAME	ZONE POINTS	ESTABLISHMENT
Uploaded video	retail_store2	4	Not assigned
CAISSE			Not assigned

N - Nano

Best for rapid iteration

Fastest setup for quick validation and lightweight machines.



S - Small

Good default for development

Balanced choice when you want better detection quality without a large hit.



M - Medium

Solid for denser queues

More accurate across crowded scenes with a moderate compute budget.



L - Large

Heavier runtime cost

Higher accuracy for harder scenes when your GPU budget allows it.



X - X-Large

Slowest option

Maximum accuracy target for controlled, powerful environments.



BackContinue to Review

Figure 5.10: Model selection screen used to choose the YOLO size before launching the feed.

The staged feed or batch of feeds is then confirmed in the review and launch panel presented in Figure 5.11.

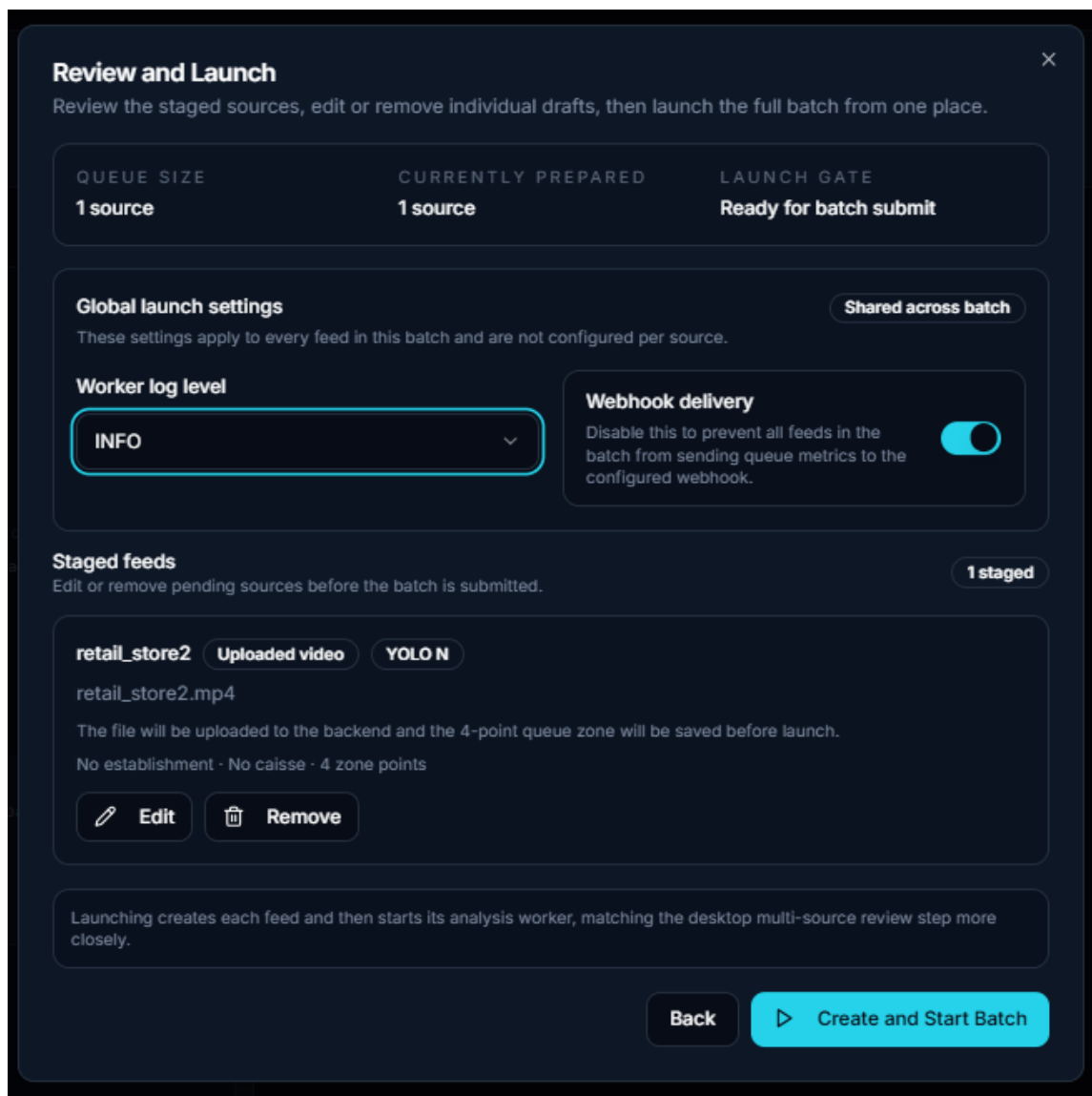


Figure 5.11: Review and launch panel used to confirm staged feeds before batch creation or launch.

5.5.9 Analytics and Historical Reporting

The analytics view extends the dashboard from live supervision into historical reporting. It aggregates archived alert history and queue metrics into a summary that helps managers identify trend shifts and recurring bottlenecks without leaving the web application.

Four KPI cards provide a fast overview: average wait time, peak queue length, stability score, and total alerts across the selected range. A period selector (hourly, daily,

weekly) controls the aggregation granularity, while date filters allow managers to isolate specific operating windows. The page then visualizes wait time and queue size trends, highlights the top-performing and underperforming zones, and summarizes the alert mix by severity.

The analytics view also supports CSV export so the same statistics can be used in external reporting. When the archive is empty, the interface seeds a mock dataset derived from active feeds to keep the dashboard informative during early pilots. Figure 5.12 presents the consolidated analytics dashboard.

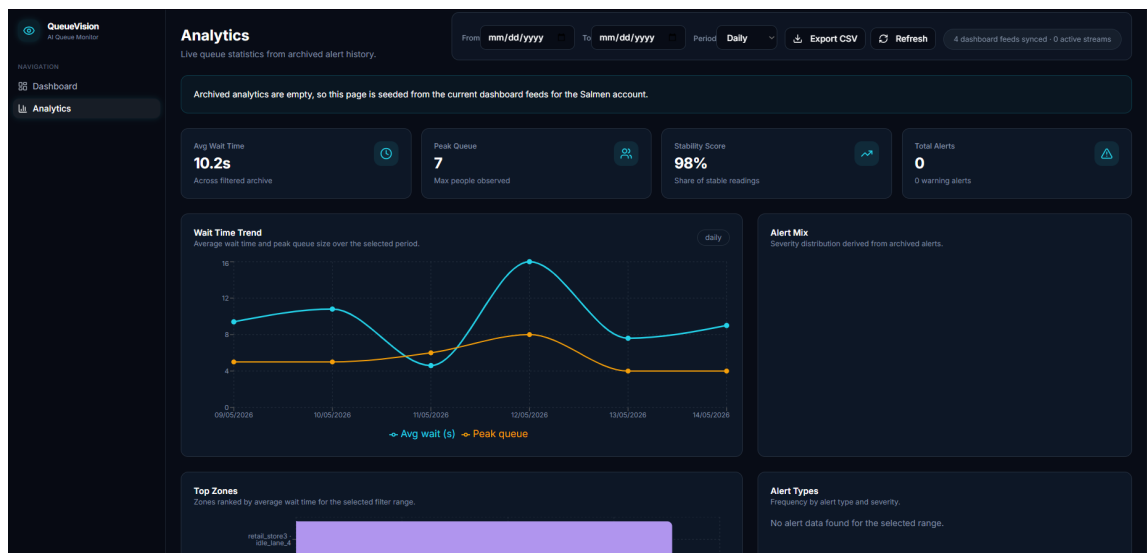


Figure 5.12: Analytics dashboard with KPI summaries, trend charts, zone breakdowns, and export controls.

5.5.10 Integration Testing

Frontend integration behavior is validated through scenario-based tests that cover dashboard setup, live event handling, transport fallback, feed routing, and error behavior.

The tests are useful not just for regression detection, but also for documenting what the sprint is supposed to guarantee. They show that the implementation is not finished when the screen renders; it is finished when the system still behaves correctly under reconnects, route changes, and fallback transitions.

5.6 Conclusion

By the end of Sprint 3, the dashboard could react to live state, switch transport modes, and stay usable when the live connection failed. That completed the frontend delivery and prepared the project for alert automation and final evaluation.

5.7 Sprint Retrospective

This sprint confirmed that separating the design foundation from live integration was the right approach. The most useful outcome was the orchestration layer that connects session state, feed actions, live metrics, and transport fallback into one predictable operator experience.

The main tradeoff was complexity at the boundary between frontend and backend. More logic moved into the dashboard, but that logic was necessary to keep the system resilient when transport conditions changed. In the final review, this was considered acceptable because the result is easier to operate and easier to explain in the report.

Sprint 4: Automation, Evaluation, and Optimization

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6.1 Introduction

Sprint 4 closed the project by connecting the queue metrics to external notification workflows and then checking that the full system still held together under realistic conditions. The sprint had two goals: route alerts from the backend through n8n to Telegram so operational staff could be notified automatically, and run a structured evaluation pass to confirm that perception, the API, the dashboard, and the alerting path still worked correctly when exercised together.

This chapter sits at the boundary between analysis and operations. Its role is to keep the runtime pipeline intact while handing alert delivery to a dedicated workflow engine, and then to validate the result.

6.2 Sprint Objectives

This sprint was organized around automation and evaluation outcomes:

Automation objectives:

- define a stable webhook payload and validate it before dispatch,
- add retry-based webhook delivery with logging for failures and successes,
- connect runtime threshold detection to alert triggering,
- configure n8n routing for authorization, cooldown, formatting, and delivery.

Evaluation objectives:

- bring backend and frontend automated tests together into one verification pass,
- confirm feed lifecycle behavior, stream continuity, and API contracts,
- improve runtime throughput and responsiveness,
- strengthen failure handling for network drops, stale data, and worker exits.

6.3 Sprint Backlog

The Sprint 4 backlog separates manager-facing user stories from technical implementation tasks. User stories represent stakeholder needs and value; technical tasks are internal infrastructure concerns.

6.3.1 User Stories

The user stories in Sprint 4 represent manager needs for automated alerting and reliable notification delivery, as shown in Table 6.1.

User StoryID	User Story	TaskID	Task
US4.1	As a manager, I want alert payloads dispatched to n8n so that operational staff can be notified automatically.	T4.1	Define the webhook payload schema and validate it before dispatch.
US4.3	As a manager, I want alerts routed through Telegram so that I receive notifications on a familiar channel.	T4.3	Create and configure the Telegram bot, store its credential in n8n, and configure the workflow.

Table 6.1: Sprint 4 user stories

6.3.2 Technical Tasks

The technical tasks in Sprint 4 represent internal implementation concerns for alert delivery reliability, workflow validation, and system-level verification.

6.4 Design

6.4.1 Alert Workflow Diagram

The Sprint 4 n8n workflow used to route queue alerts through cooldown checks, archival, and Telegram delivery is illustrated in Figure 6.1.

Task ID	Task	Subtask ID	Subtask
T4.2	Add retry logic, timeout handling, and logging for webhook delivery to ensure notifications are not silently lost.		Implement delivery resilience with exponential backoff and failure tracking.
T4.4	Prepare PowerShell scripts for single-alert, multi-feed, and execution polling scenarios to validate n8n workflow behavior.		Create repeatable test automation for workflow routing and alerting paths.
T4.5	Run backend and frontend tests together and confirm lifecycle, streaming, and API contracts under realistic conditions.		Execute comprehensive integration verification to catch issues before delivery.
T4.6	Apply runtime tuning: configurable image size, frame stride, async dispatch, and playback pacing to improve processing responsiveness.		Implement performance optimizations for the main processing loop.
T4.7	Improve feed worker supervision, frame handoff continuity, and websocket reconnection to strengthen fault tolerance.		Enhance system recovery from transient failures and network disruptions.

Table 6.2: Sprint 4 technical tasks

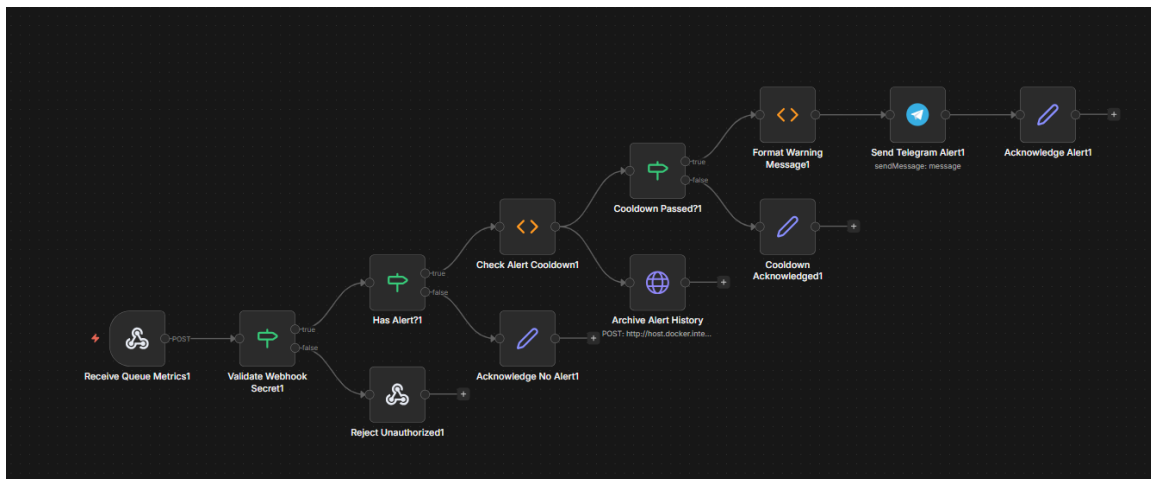


Figure 6.1: Actual Sprint 4 n8n workflow automation diagram used to route queue alerts through cooldown checks, archival, and Telegram delivery.

The automation chain includes the runtime metrics collector, the webhook serializer, the HTTP delivery client, the n8n workflow, and the final notification channel. When an alert needs to be preserved for later review, it goes back to the backend through a dedicated archive endpoint. The workflow checks the webhook secret, applies the cooldown gate, formats the alert, and either acknowledges the payload or forwards it to Telegram.

6.4.2 Evaluation and Resilience Activity Diagram

The evaluation and resilience path is summarized in Figure 6.2. This diagram complements the n8n workflow by covering the verification, tuning, and fault-tolerance backlog items that are not part of the alert-routing diagram itself.

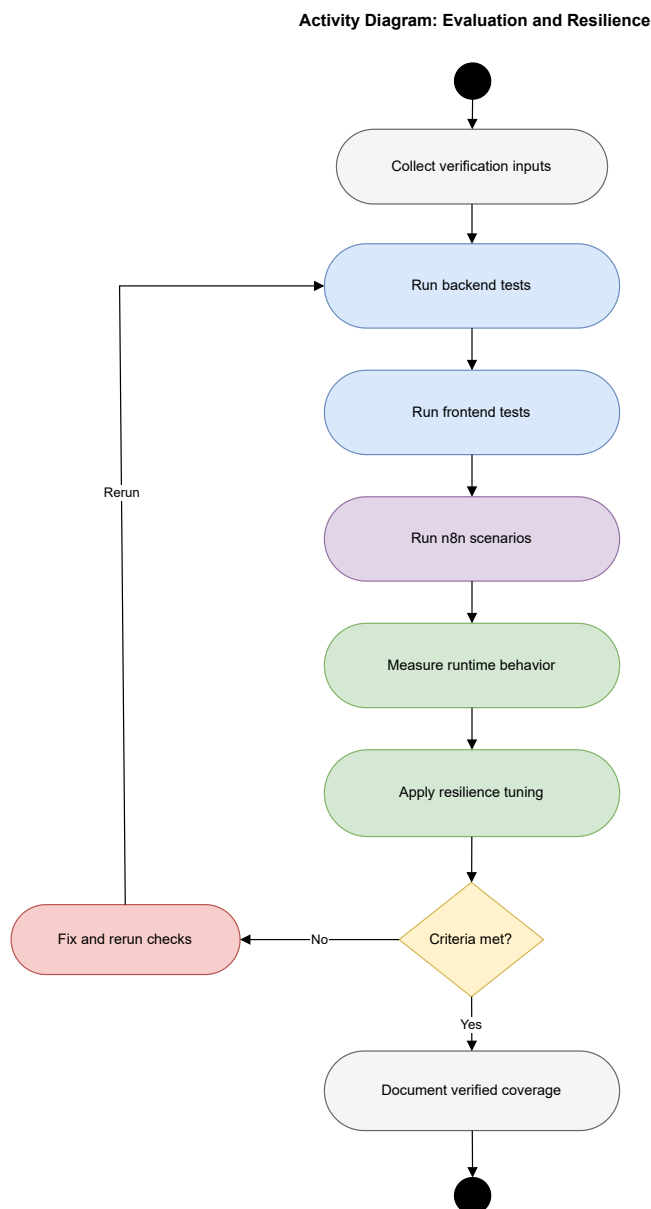


Figure 6.2: Sprint 4 evaluation and resilience activity diagram for tests, tuning, and recovery verification.

The activity starts by collecting the verification inputs, then runs backend, frontend, and n8n scenario checks. After that, runtime behavior is measured, tuning and recovery

improvements are applied, and failed cases loop back into the affected test layer until the acceptance criteria are met.

6.4.3 Verification Plan

Validation followed a layered strategy that covered the system from the backend to the dashboard:

- **Backend unit/integration tests:** queue analytics logic, API endpoints, runtime streaming behavior, and webhook payload handling.
- **Frontend integration tests:** dashboard setup flow, websocket event handling, media fallback behavior, and API error handling.
- **Scenario tests:** n8n send/acknowledge flows, unauthorized access paths, and multi-feed alert scenarios.

6.4.4 Performance Targets

The optimization work was guided by the main operational requirements of the project:

- processing responsiveness, especially loop cadence and processed-frame throughput,
- stream continuity when a feed changes state,
- webhook impact on the main-loop latency,
- dashboard freshness when websocket quality degrades.

These targets intentionally mirror the behavior that an operator would notice first, so the evaluation reads as a system-level check rather than a narrow benchmark exercise.

6.5 Implementation

6.5.1 Alert Payload Structure

The webhook schema is centered on `QueuePayload`, which keeps the information needed by n8n in one structure. It carries:

- source metadata (timestamp, frame, source, feed ID),
- queue metrics (λ , μ , wait time),
- alert flags and severity fields.

Before anything is sent, the payload is validated so malformed records do not leave the backend.

6.5.2 Delivery Reliability

To reduce the risk of missed notifications, the delivery client uses a timeout, a retry policy, an optional shared-secret header, and logging for both successes and failures. Historical payloads remain in SQLite, which gives the system one place to recover past events if needed.

6.5.3 Telegram Bot Setup

Before the workflow could send operational alerts, a Telegram bot was created and connected to the project as the final notification channel [31]. Its API token was added to n8n as a credential instead of being hardcoded, which kept the workflow portable and prevented secret exposure in the exported JSON. Once the credential was available, the Telegram node could use it after the webhook validation and cooldown checks passed.

6.5.4 Alert Dispatch in the Runtime Loop

Inside the main loop, alert sending is moved to `AsyncWebhookDispatcher` so the perception pipeline keeps running without waiting on the network. This makes it possible to send periodic metric updates and immediate alerts, while a deduplication window limits repeated notifications for the same situation.

6.5.5 n8n Workflow Routing and Telegram Delivery

The n8n alert workflow follows a simple path [17]:

- webhook reception and secret validation,
- alert detection and cooldown gating,
- alert history archival via backend API,
- message formatting and Telegram send node,
- explicit acknowledge/reject branches.

Keeping the no-alert, alert-suppressed, and alert-delivered branches separate makes the behavior easier to audit later.

6.5.6 Workflow Validation

Workflow validation was supported by PowerShell scripts for single-alert checks, automated execution polling, and multi-feed scenarios. These scripts made it possible to replay webhook submissions and confirm that each branch behaved as expected during sprint verification.

6.5.7 Verification Coverage

The test suite covers API transport behavior, runtime stream handling, webhook delivery, dashboard integration, live-state synchronization, API error cases, and transport

fallback behavior. Taken together, these tests check the API envelope, the event contract, stream handoff, and the setup workflow.

6.5.8 Runtime Tuning

The main tuning actions applied across the runtime loop and configuration layers included:

- configurable detector image size and frame stride,
- explicit control over the inference device and its diagnostics,
- asynchronous webhook dispatch so the processing loop stayed responsive,
- real-time playback pacing for file-based sources.

6.5.9 Fault-Tolerance Improvements

Runtime resilience was improved at several levels:

- feed worker supervision and exit recovery in `FeedRegistry`,
- capture-to-worker frame handoff continuity in `FeedFrameStreamManager`,
- websocket reconnect and fallback behavior on the frontend side.

6.5.10 Overall Outcome

By the end of the sprint, the system had reached a stable integration point where perception, analytics, the API runtime, the dashboard, and n8n automation operated as one pipeline. What remains is mainly deployment hardening and longer-term benchmarking, not missing core functionality.

6.6 Conclusion

Sprint 4 closed the project with operational alert routing and structured testing. The automation path now delivers alerts from the backend to Telegram through n8n, while the evaluation pass confirmed that the full stack still works under realistic conditions. At this stage, the platform is ready for demonstration and academic evaluation, while the remaining work is mainly deployment hardening and longer-term benchmarking.

6.7 Sprint Retrospective

This final sprint showed that the most important delivery was not any individual feature but the end-to-end integration. The automation work gave the system its last operational channel, and the evaluation work confirmed that the earlier sprints still held up when tested together.

The main lesson is that alerting and verification are not secondary tasks. Without reliable dispatch and structured testing, the earlier perception and dashboard work would remain incomplete from an operational perspective.

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7.1 Summary

This end-of-studies project produced a complete queue wait-time estimation platform that brings together computer vision, queue analytics, web supervision, and automation workflows.

In practical terms, the system now includes:

- a perception pipeline based on YOLO detection and ByteTrack identity continuity,
- queue-zone filtering and event-driven estimation of λ , μ , and expected wait time,
- adaptive wait-time outputs that respond to queue conditions,
- a secure authentication layer for manager sign-up, sign-in, session refresh, and administrator account governance,
- a typed FastAPI backend-for-frontend with feed lifecycle control and live web-socket events,
- a React dashboard for monitoring, setup, and runtime interaction,
- n8n and Telegram integration for operational alert routing.

The resulting architecture shows that a full monitoring-to-action loop can be built from modular components without losing maintainability or testability.

Across the four consolidated sprint chapters, each sprint adds one layer to that loop: Sprint 1 establishes perception and queue analytics, Sprint 2 builds the backend services and API integration, Sprint 3 delivers the web dashboard and live monitoring, and Sprint 4 completes the automation pipeline and validates the full system.

7.2 Objective Review

The initial objectives were achieved at the project scope level:

1. **Real-time detection and tracking:** achieved through dedicated detector and tracker modules.
2. **Queue-zone counting:** achieved with polygon-zone management and in-zone identity filtering.
3. **Queue metric estimation:** achieved for arrival rate, service rate, and wait time.
4. **Adaptive wait-time estimation:** achieved through automatic formula switching based on queue load.
5. **Web monitoring:** achieved through a multi-feed dashboard with live updates.
6. **Automation integration:** achieved through webhook transport, n8n routing, and Telegram notifications.

The main remaining gap is not architectural completeness, but the depth of long-horizon evaluation under broader operational scenarios.

7.3 Future Work

Possible extensions include:

- multi-zone and multi-server queue modeling for parallel cashier configurations,
- richer alert strategies with cooldown tuning, escalation policies, and acknowledgment loops,
- edge deployment optimization through ONNX/TensorRT profiling and resource-aware scheduling,
- long-duration field evaluation with domain-specific KPI baselines,
- stronger security controls for production webhook exposure and secrets management.



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